AWI–ESM3 v3.5 PI-control tuning campaign — summary

Fifty-eight rounds, from a boreal forest that vanished to a CMIP7 spin-up on tagged components, by way of a Labrador Sea traced to its GM setting, a Southern Ocean cap that no lever strengthened, and a deep ocean that drifts for as long as it runs

OpenIFS 48r1 / FESOM2 / LPJ-GUESS / OASIS / XIOS, TCO95L91 / CORE3

supersedes the 2026–08–24 summary; rounds 41–46 added 2026–10–03, rounds 47–58 2026–10–15

2026–10–15

Abstract

The short reading of a campaign whose full record runs to 183 pages (`report.tex`). Three objectives throughout: the **global radiative imbalance**, the **Southern Ocean**, and the **NH high-latitude cold bias / boreal forest**. §4 is the whole campaign in the order it happened, rounds 1 to 58; the sections after it are the state now.

Where it ended (2026–10–15). The tuning is closed and the CMIP7 spin-up runs: **AWI-ESM3-CC-LR-CMIP7-Spinup**, the carbon-cycle variant on tagged components (FESOM 2.8.2, OpenIFS 48r1v6), from PHC3, with `cvmix_TKE` alone, hemispheric GM, `albsn` 0.82 and icebergs from the model's own mass budget, at 41 simulated years per day. After 160 years it is 0.4–0.5 K colder than the tuning line at the same age, its deep ocean drifts at half the tuning line's late rate, and ocean and land together lose 0.41 $\mathrm{GtC\,yr^{-1}}$ against a target of 0.1, which decides when it ends (§4.19).

The founding finding still stands. Every lever that lowers the TOA imbalance does it by cooling the model. The 29 early runs lie on a line, not in a cloud: about 0.6 K of global cold per $\mathrm{W\,m^{-2}}$ of TOA. Underneath that line sits a structural floor that no atmospheric retune reaches — a boreal forest that LPJ-GUESS does not deliver, worth −3.1 K of Siberian DJF on a clean with/without pair.

The campaign left lever-screening behind in September and produced a single 160-year pre-industrial line, **PICAL_ccnice** (1940–2100), on the v3.5 defaults. It is not converged, and the drift is cold: 0.214 ± 0.025 K/decade globally over 2060–2099, 0.511 ± 0.118 at 60–90 N, with March NH ice extent growing.

Positive TOA does not mean warming here. Net TOA over 2080–2099 is $+0.167\,\mathrm{W\,m^{-2}}$ while the surface cools. Total ocean heat content is flat (+0.04 ZJ/decade): the upper 700 m loses to below 700 m at $0.32\,\mathrm{W\,m^{-2}}$ equivalent, almost exactly balanced. **The cold drift is an ocean redistribution, not an energy leak.** On one axis from 1850 to 2139 the warm layer at 300–1000 m builds without interruption, and of the mixing schemes it is IDEMIX that feeds 700–1500 m (+31 mK/decade against +9 for TKE alone and +11 for KPP). A TKE-only continuation is the first run aimed at it.

The Southern Ocean has resisted the whole campaign. It generates the imbalance rather than responding to it (+0.383 and $+0.353\,\mathrm{W\,m^{-2}}$ from the two southern bands), 84 % of its cloud error is already present in AMIP over a correct ocean, and five more levers were excluded in round 37. It sits at 62 % of observed winter ice volume and the ACC at 98.9 Sv against 134 ± 11.2 observed. Its surface cap is too weak in every run, 78 % of observed in the open ocean and 31 % in the Weddell region, and none of five freshwater and sea-ice levers tried in October strengthened it.

The boreal forest was gone, and the line now carries one. BNS 0.001 and 5 % tree cover in the Siberian box at 2099 on ccnice, every establishment gate open. **PICAL_crunveg** (on albedo since 2100) restarts LPJ-GUESS from the CRUNCEP state with the merged canopy code and keeps 81 % of its larch after 19 years; the forest costs about $2 \times 10^3\,\mathrm{km^3}$ of NH ice and +0.24 K globally.

The GM regime, a pre-industrial reference, and the Antarctic pack. The Labrador Sea ice-capped regime appears only under Round 36's GM 2500 / Redi 1000 (61 runs); GM 2500's heat-uptake benefit is Southern Ocean ventilation at 34–55 S. Of four 20-year GM arms at the production step, **hemispheric GM** (2500 south, 1000 north) restores Labrador convection and moves nothing else significantly; it is adopted and overshoots. **Against a pre-industrial reference the model is +0.46 K too warm**, in summer mid-latitudes, with a cold Arctic winter: too warm and a positive imbalance now point the same way. The Antarctic winter warm ring sits over a too-diffuse pack whose drift over-responds to winds that are about right.

1 Scorecard against the three objectives

objective	at the start	what the campaign did to it	where it stands
Radiative imbalance	Coupled net TOA +1.44 to +1.73 W m^{-2}, and every lever that lowered it cooled the model 0.6 K per W m^{-2}.	Measured that ≈45 % is atmospheric (AMIP +0.67); rebased the target to 0; found that decade-4 TOA follows mid-depth heat burial (r = +0.72) and that GM is the lever that reaches it.	**Partly understood, not fixed.** gmhemi1800 2130–39: net TOA +0.52, surface +0.67, ocean uptake 0.69 W m^{-2} mostly below 700 m, while the line still cools 0.06 K/decade. Against a PI reference the model is +0.46 K *warm*, so the imbalance and the temperature error now point the same way (§10). The CMIP7 spin-up sits at +0.55 after 135 years, 0.5 K colder than the tuning line at the same age.
Southern Ocean	Largest single regional term, +7.46 W m^{-2} period-clean, and a winter over-ice warm bias.	Established it is cloud *amount* at 60–45 S and *opacity* at 90–60 S; that 84 % is inherited from AMIP; traced and priced the marine CCN path; excluded nine levers in total.	**Open; to be documented as structural.** 67 % of observed winter volume (gmhemi1800), ACC 30 % weak, GM 2500 costs 16.7 Sv. The winter warm ring is a diffuse pack, not a missing edge, and its drift over-responds (§11). The surface cap is 78 % of observed (31 % Weddell) and five freshwater and ice levers left it there; icebergs are on in the spin-up.
NH cold bias / boreal forest	Siberian JJA −2.58 K against ERA5; `cvh` 0.63 → 0.02 when LPJ-GUESS is switched on.	Split it into a universal part, an orographic part and a boreal-specific part; recovered ≈1.0 K of the last with G4/K1 and another 0.77 with the stable-BL lever; closed land albedo and surface exchange as explanations.	**Atmospheric half improved; the forest is back on the line.** crunveg keeps 81 % of its larch; against a PI reference the NH summer land is now *warm* (+2.1 K at 30–60 N) and the Arctic winter cold (§12, §10). The spin-up lost forest in its first decades (60–90 N tree fraction 9→6 %), the likely cause of its colder state.

2 Where the model stands

PICAL_ccnice 2090–2099, **gmhemi1800** 2130–2139 (the crunveg line with hemispheric GM, §9) and the CMIP7 spin-up 1975–1984 unless stated. Observations as noted.

quantity	ccnice	gmhemi1800	spin-up	observed	source
net TOA (ccnice 2080–99)	+0.167	+0.519	+0.55	0 for a control	—
net surface (ccnice 2080–99)	+0.357	+0.673	—	0 for a control	—
T2m global trend, K/dec	−0.214	−0.059	+0.04	0 for a control	2060–99; crunveg line 2101–39
T2m global, PI bias	—	+0.46 K	—	0	ERA5 + HadCRUT5
NH April ice volume	34.56	31.69	—	31.19	GIOMAS 1989–99
NH September ice volume	13.97	12.18	16.1	11.96	GIOMAS 1989–99
SH September ice volume	12.33	13.33	14.5	19.89	GIOMAS 1989–99
SH February ice volume	0.73	1.33	1.9	2.07	GIOMAS 1989–99
Labrador March ice, box	—	0.10	—	0.16	HadISST2 1979–2008
Drake Passage transport	98.9 Sv	—	—	134 ± 11.2	Cunningham 2003
				173.3 ± 10.7	Donohue 2016
cavity basal melt	1062 Gt yr^{-1}	—	—	∼1100	Adusumilli 2020
				∼1500	Rignot 2013

quantity	ccnice	gmhemi1800	spin-up	observed	source
CMPI (global)	0.903	0.861	0.918	lower is better	2125–2139 for gmhemi1800
Siberian tree cover (FPC)	0.050	—	—	0.415	offline CRUNCEP

Volumes in $10^3\,\mathrm{km}^3$. The NH is expected *above* a satellite-era record for a pre-industrial run; the SH is close to like-for-like because GIOMAS has no significant Antarctic trend across it. The surface imbalance includes the snow-enthalpy term (§16). gmhemi1800 is not on the same years as ccnice, so the columns differ by land state, GM and 40 years of drift together. The spin-up column is a different ocean state altogether: 135 years from PHC3 on the final settings (CMPI 0.918 against the tuning line's 0.927 at the same age).

3 What v3.5 carries, and where each number came from

The tuning now lives in `configs/setups/awiesm3/awiesm3.yaml` rather than in runscripts, so a fresh install reproduces the line. Everything below is namelist-only except where noted, which is what makes it mergeable with EC-Earth.

parameter	value	why, and which round decided it
`RCL_INPSEA`	0.1	D2b, round 11. Reduced marine ice nuclei → brighter supercooled Southern Ocean cloud.
`RCL_INPPMIN`	70000	Round 28 adopted 50000 (S4) and round 38 retired it: the benefit reverses sign coupled. 70000 follows `sucldp.F90`'s own recommendation.
`RSBLB`	2.0	Rounds 31–32. The winter land lever; Arctic DJF screen +0.614 K coupled at net TOA −0.001.
`RCCNSEAICE`	15.0	Round 38. Open-water weighting of the whitecap sea-salt source, which OpenIFS otherwise evaluates over sea ice. Gives the current line its name.
`RVRSMIN(3,4)`	1000	Round 14 (G4). A cap set by winter damage, not a saturation knee.
`ECE_SNOW_SCF`	3 (+9 coeffs)	Round 20 (P3/P5), fitted to Russian snow-course observations after the `tanh` form was falsified against 174 676 station soil records.
`RVVEGALB`	40 values	Round 17 (K1). Snow-free land albedo corrected against CERES; closed land albedo as an explanation.
`cavity_gamma_scale`	0.6	Round 37. 1190 $\mathrm{Gt\,yr^{-1}}$ against 1893 uncalibrated, 1100–1325 observed. FESOM PR #1064, merged.
`K_GM_max` / `Redi_Kmax`	2500 / 1000	Round 36 (16E). Lowest TOA plateau kept. §8 is the bill nobody scored, and the Labrador capping another (§9).
`K_GM_max_NH` / `K_GM_hemi_trans`	1000 / 10	Round 43 (gmhemi1800). 2500 stays in the SH, 1000 in the NH, linear over 10° across the equator. A regional gate, accepted explicitly after consulting the FESOM core developers; FESOM PR #1112, in the 2.8.1 tag. Effective values 980 south and 390 in the Labrador Sea (round 51).
`mix_scheme`	`cvmix_TKE`	Round 39 adopted TKE + IDEMIX; rounds 46–47 dropped IDEMIX, which feeds the mid-depth warming, at the cost of a slowly eroding Antarctic summer pack.
icebergs	on	Rounds 53, 57. (calving − cavity basal melt) released as icebergs, mapper calving off for Antarctica; grounded bergs drift slowly (`l_allowgrounding` 1) and iceberg cooling is capped at −1.85 °C since the 1902 crash (round 58).
sea-level pressure to FESOM	on (`-cc`)	Round 49, Bug XIII: REcoM's gas exchange needs it. FESOM 2.8.2, OpenIFS 48r1v6.
sea-ice roughness	released	Round 55. Exposed in `NAMSURFTUNE`, left at the defaults: the atmosphere gives the stress back.
`use_momix`	false	Round 38; the production line's first branch tests it.
`albsn` / `albsnm`	0.82 / 0.65	Round 38 computed 0.80 from the model's own melt seasons; round 47 raised it to 0.82 to stop the surface warming of the TKE-only line (0.83 ran away in the Arctic). FESOM ships 0.81/0.77, so these are deliberate. `albi`/`albim` stay at FESOM's 0.70/0.68 — darkening them was falsified, they are pond-free.

Not levers at all: `h0min`/`h0max` are hard-coded (0.5/1.5) in the coupled `ice_thermo_cpl.F90`, and the namelist `h0`/`h0_s` are read only by the uncoupled thermodynamics, which is compiled out (round 45).

Deliberately unset: `RSNOWLIN2` stays at the code default 0.030. Raising it to 0.04 is the only lever in fifty arms that moves the tropical longwave deficit (+2.74 W m^{-2}, 120–127 % of it) and it warms the Southern Ocean too — but it costs +0.42 W m^{-2} globally. It remains the campaign's most interesting unadopted parameter.

4 The campaign in the order it happened

One paragraph per phase, with the number that settled it. The full record of each is in `report.tex`, whose §1 carries the same order.

4.1 The trade-off, and the floor underneath it (2026–07, rounds 1–8)

Twenty-nine coupled runs, screened on net TOA against the ERA5 temperature bias, produced the campaign's founding picture: **the runs lie on a line, not in a cloud**. Every W m^{-2} of TOA improvement cost about 0.6 K of global cold, from 080a at $(1.73, +0.28)$ to 10A at $(0.57, -0.93)$; the coupling and ocean-mixing family cut the imbalance 30–57 % but pushed the planet 1.7–2.1 K too cold. Centred RMSD was 1.41–1.64 K across all 29 runs, a ±8 % spread against a raw-RMSD spread of 64 %: **the runs differ by an offset, not by a pattern**. That identified a shared structural floor rather than a tuning deficiency, and sent the campaign looking for a warming lever instead of another retune.

4.2 The floor is LPJ-GUESS, on a clean pair (2026–07–14)

A purpose-built no-LPJG reference — same CORE3 ocean, same TCO95 atmosphere, matched branches, HTESSEL with prescribed satellite vegetation, 30 years — differs from the baseline in LPJ-GUESS alone. **Siberian high-vegetation cover `cvh` goes** 0.63 → 0.02 when LPJ-GUESS is switched on (pan-boreal 0.49 → 0.07), and Siberia is −3.1 K colder in DJF. The no-LPJG run also beats the baseline on global T2m RMSD (1.514 against 1.717 K) and on CMPI (0.738 against 0.745) at essentially unchanged imbalance, so the warming lever is real and nearly free. The cold effect is present in every month except high summer, which already ruled out snow-albedo masking as the whole mechanism: that is a sunlight effect and cannot explain an equally cold polar night.

4.3 The feedback-free AMIP reference (2026–07–27/28)

An AMIP testbed was built so the atmosphere could be scored with the vegetation–albedo feedback open but the coupled runaway severed. **(i) About 45 % of the imbalance is atmospheric**: AMIP net TOA is +0.67 W m^{-2} against a coupled +1.44, and with SST prescribed that cannot be ocean heat uptake — the best branch at the time sat only 0.25 W m^{-2} above that floor, so the Part-I lever family was nearly exhausted. **(ii) The boreal *summer* cold bias is atmospheric too**: AMIP reproduces essentially all of the coupled Siberian JJA bias (−2.16 against −2.19 K) while its *annual* bias is only −0.89 against −3.0 coupled. Summer became an atmospheric problem and September–April stayed the coupled/vegetation one.

4.4 Rounds 10–12: a noise floor, a dead namelist, and the best boreal lever (2026–07–30→08–02)

Three infrastructure findings changed how everything afterwards was measured. The **4-year window was resolving nothing** in the boreal, so arms went to 46 years and a detection threshold became mandatory. The **period offset** was measured rather than estimated (+0.42 K for Siberian JJA, against a 1.12 K guess that had already been used to withdraw a conclusion). And **`NCMIPFIXYR` sat in a dead `&NAMECECMIP6` block**: the AMIP runs were never at 1850 forcing, GHG ran 1869→1911, and Krakatoa is visible in 1883–86 in every arm. Deltas were unaffected, the absolute baseline was offset ≈0.15 W m^{-2}, and the coupled runs were correctly pinned. On the science side D2b (ice nuclei) was adopted for the Southern Ocean and F4 — boreal surface exchange — became the best lever to that point, with G1 = F4 + D2b the best configuration.

4.5 Where the bias is not: the vertical structure (2026–08–02)

Before spending more levers, the column was decomposed. The Siberian cold bias is **confined below 850 hPa**; **the soil and skin are the least-biased parts of the column**; and **all 20 surface types are cold in every season**, so it is not a vegetation-tile problem. That decomposition eventually retired the surface-side lever family as a class, and it split the boreal budget into ≈0.7 K universal, ≈0.3 orographic and 1.3–1.6 boreal-specific — with G4 taking ≈0.95 of the last, and the remainder not reachable by a boreal-indexed lever at all.

4.6 Rounds 13–17: land albedo, and half of it is the snow tile (2026–08–03→05)

Half the Siberian shortwave deficit turned out not to be cloud. Of a +0.0154 land albedo excess against CERES, +0.0074 is the snow tile and +0.0080 the snow-free surface; every forest type is inside +0.006 once snow is masked per cell per month. The signature is **snow-melt timing** — accumulation is right, the melt is a month late and 30 % too slow, and May is a runaway. Round 14's G4 was adopted (Siberian JJA +0.952 K, $t = 7.68$, SO SW RMSE 6.88 → 4.80) with `RVRSMIN` capped at 1000 by winter damage, and round 17 adopted K1 = G4 + the snow-free albedo correction (+1.036 K, global T2m RMSE 1.579 → 1.524, no Arctic energy penalty). Rounds 13 and 16 — snow cover fraction and skin conductivity — were both rejected on their own predictions.

And then land albedo was closed as an explanation. Removing 74 % of a directly measured albedo bias bought 0.089 K of a ≈0.7 K target; a perfect fix buys ≈0.12 K, one sixth. That is the shape of most of this campaign: the diagnosis was right, the leverage was not there.

4.7 Rounds 18–21: the snow scheme, falsified on observations and refitted (2026–08–05→07)

The strongest summer lever built (I3, +1.322 K Siberian JJA, closing the surface SW deficit to CERES exactly) was *not* adopted, because it cost +1.72 W m^{-2} at 60–90 N. The sub-grid snow depletion scheme that HTESSEL lacks was implemented, validated offline, and it worked — melt-out 12 May against a satellite 11 May — and it cost −16.2 K of coupled DJF soil for +1.2 K of summer. The `tanh` functional form was then **falsified against observations rather than preference**: its range is open at 1, so complete cover is unrepresentable, while Russian snow courses report exactly 10/10 in 99.74 % of 14 369 Siberian DJF surveys, and the scheme costs −20.2 K against 174 676 station soil observations where the scheme *off* is right to +0.8 K. The damage was distributional, not a matter of mean cover: DJF cover is identical to 0.002 across the arms while the soil differs by 23 K. P3, refitted to observations, was adopted; P5 followed. **Direct station data (RIHMI-WDC) entered the campaign here**, and the campaign stopped depending on ERA5 — an HTESSEL sibling — for any soil claim.

4.8 The forest fails by two routes (2026–08–05)

BNE/BINE are *gate-excluded*: 0.7 % of NE-Siberian cells clear `tcmin_est`= −30 and their LAI is exactly 0.000. BNS (larch) has *no* cold gate, clears its GDD gate on 84 % of cells, and is *outcompeted* — grass carries 5.8× its LAI. Since NE Siberia is ≈80 % larch, **the winter-gate mechanism does not explain the region that matters**. This distinction has been rediscovered several times since and is why §12 states it explicitly.

Round 27 then measured the transfer function that the campaign's founding premise assumed: offline 2000-year LPJ-GUESS equilibria give TREE 0.202 / BNS 0.034 under AMIP forcing, 0.261/0.058 at +1 K, 0.327/0.099 at +2 K, against 0.415/0.227 under observed forcing. Monotonic, not saturating, and +2 K recovers 59 % of the gap — **closing the cold bias should recover most of the forest**. A separate transfer test showed **forcing alone halves boreal tree cover**: same parameters, same 10 043 gridcells, `global.ins` byte-identical, Siberian TREEFPC 0.357 → 0.163 (−54 %) when the spin-up forcing is swapped from CRUNCEP to what the model delivers.

4.9 Rounds 22–23: the energy target re-based, and the Southern Ocean promoted (2026–08–09)

Comparing a PI run against a 2000s CERES window conflates the climate difference with the bias. Measured period-clean, **the radiative error is +1.23 W m^{-2}, about twice what the PI framing reported**. The tropical deficit largely dissolved (−2.51 PI-vs-CERES against −0.67 period-clean); the Southern Ocean got *larger* (+7.46 W m^{-2}, +0.735 of the global mean) and became unambiguously the largest single regional term. Its deficit is **cloud amount, 65.5 % of it**, zonally uniform to 1.42 pp — and no lever in the campaign moves cloud area, the best managing +0.63 pp of a 6.43 pp gap. The reference question was settled too: CERES and MODIS put SO cloud area at 89.5 and 89.3 % against the model's 83.1, while **ERA5 reproduces the model's bias because ERA5 is IFS** and is therefore not an independent cloud reference. Meanwhile the global cold bias was located at the tropopause, peaking at 200 hPa over both poles, with the atmosphere cold *and* dry (−1.62 K and −10.1 % over 700–300 hPa) in a pair that cancels in OLR.

4.10 Rounds 24–26: marine biogenic CCN, traced end to end and priced out (2026–08–09→10)

The one lever that reached the Southern Ocean in the right season. Droplet number there is **correct in the mean** ($N_d \approx 73\,\mathrm{cm}^{-3}$ against 60–120 observed) but **inverted in phase**: the model is lowest in DJF because it is wind-driven, where observations peak in DJF from biogenic sulphate, and the CRE error is worst in exactly those months. A DMS emission climatology already shipped at TCO95 and nothing read it; it was wired through `ocp-tool`, defaulted off, and screened. At $S = 166$ it moves SO SW CRE $-7.25\,\mathrm{W\,m}^{-2}$ in DJF, 67 % of the deficit in the worst season, from the bloom rather than a tuned constant. **It still cannot be adopted**: the tropical cost is $0.0129\,\mathrm{W\,m}^{-2}$ per unit S, so the tolerance binds at $S \approx 39$, below the published floor of 43, and scaling cannot separate the bands because tropical background CCN is *lower* than the Southern Ocean's.

4.11 Rounds 27–28: a tuned stack nobody ran, and S4 (2026–08–10→11)

A jointly-tuned eight-parameter cloud stack for TCO95L91 was found sitting unused in the v3.4.2 DARS2 runscripts, with no `fort.4` on disk setting any of it; it had been invisible because `preflight` looked in one place. Taken apart, it yielded `RSNOWLIN2`, still the only lever that moves the tropical longwave. The round also established that **Southern Ocean levers must be scored on net, not SW CRE** — longwave cancels a consistent 25–29 % across the 44-year archive. Then S4 (`RCL_INPPMIN` 70000 → 50000), one namelist number that had sat unevaluated for two days, came out as the best configuration the campaign had: SO surface bias +3.66 → −0.96, global net TOA +0.64 → +0.25. **It was the headline of the 2026–08–11 summary and is now retired** — its benefit reverses sign coupled (Siberian JJA +1.194 K AMIP against −0.365 K coupled). The habit of striking falsified claims through rather than deleting them dates from this episode.

4.12 Rounds 29–30: the land-surface repair, Raupach, and an inherited bias (2026–08–13→19)

Raupach canopy roughness is dead structurally, not statistically. Its winter mechanism needs a canopy, and the model's Siberian trees are 2.89 m tall with 2 cm stems, giving $z_0 = 0.0132$ m against IFS's 0.058 — it makes the winter surface 4.4× smoother. **The lever cannot bootstrap the forest it requires.** The land-surface repair (107 new `fail()` calls classified, peat and LUH3 reconciled to one epoch) is a trade: +0.308 K of Siberian DJF soil against a JJA bias moved from −0.673 to −1.123. Round 30 then produced the reframing that governs the Southern Ocean work since: **84 % of the coupled SO cloud error is already present in AMIP** over a prescribed, correct ocean. It is not a coupling problem. Round 30 also killed the "$-1.0\,\mathrm{W\,m}^{-2}$ structural leak": it was the omitted snow-enthalpy term, and with it the column conserves to 0.02–0.04 $\mathrm{W\,m}^{-2}$ and a TOA-zero run drifts −0.02 K/century. **Tuning net TOA to zero does stop the ocean drift** — which is what makes §5 surprising.

4.13 Rounds 31–32: the stable boundary layer, and the campaign's best run (2026–08–21→23)

The decomposition said the winter land bias sits between the screen and 925 hPa: in the best arm the air-side inversion bias is +2.81 K while the skin-side bias is +0.11 K. That excluded every surface-side lever by construction and pointed at stable mixing, so `&NAMVDFS` was created to expose it. `RSBLB 2.0` gives Arctic DJF screen +0.614/+0.674 K coupled, and **it is free in energy terms coupled** — global net TOA −0.001/−0.027, both null, against AMIP's +0.175: the tropical shortwave gain is absorbed as ocean warming and cancelled in the longwave. **Winter and summer moved together for the first time in the campaign.** 11R scored CMPI 0.7577 against 11E's 0.8237, the best in the campaign. The mixing-length route (SB3) was falsified: its tropical SW cost per K of inversion is 3.43 against `RSBLB`'s 0.88, so the trade is a property of the scheme, not of the knob. Only about half the residual land bias is reachable by any mixing lever — of −2.23 K, ≈1.0 is excess inversion and ≈1.2 is a deep cold bias still present at 850 hPa.

4.14 Round 33: the Southern Ocean is two problems (2026–08–23)

It generates the imbalance rather than responding to it: 90–60 S and 60–45 S contribute +0.383 and $+0.353\,\mathrm{W\,m}^{-2}$, the two largest positive terms. The two bands need *opposite* corrections — at 90–60 S cover is right to 2 pp but each cloud is 8 % too dim, at 60–45 S there are 5 pp too few clouds each 9 % too bright — and every campaign lever raises opacity, which is why one band is now overcorrected while the other is not. Cloud phase flips sign at 60 S, and the INP family had been retired on a measurement taken in the one band where

the model is too liquid. The ice melts from above in summer and fails to refreeze from below in autumn, with a +0.9 K anomaly at exactly 100 m that autumn mixing entrains. **The ice is not too dark** (compact-ice albedo 0.760–0.778); it is too diffuse — no cell exceeds 90 % concentration in January–March, so the area-mean albedo falls to 0.49–0.51.

4.15 Rounds 34–36: where the heat goes (2026–09–06→12)

Round 34 found four defects in the coupled ice skin (§16). Round 35 asked where the heat lands: **below 300 m**, with the top 100 m gaining $+0.013\,\mathrm{W\,m^{-2}}$ while 300–2000 m gains +1.388; the tropical mixed layer is *not* too deep (43.6 against WOA18's 51.6 m); the tropical deep warming is **heave, not diapycnal flux**; and heat crosses density surfaces at the mid- and high-latitude winter outcrops where the mixed layer is +48.9 to +155.7 m too deep. The Southern Ocean *loses* heat rather than burying it, with 18.7 % of month-nodes mixing to the sea floor.

Round 36 closed the loop: **decade-4 TOA follows mid-depth heat storage** across 20 runs ($r = +0.72$), **stronger GM is the lever that reaches it** (GM 1500 cuts first-decade 700–2000 m uptake by ≈30 %), and 16E (`K_GM_max` 2500, Redi 1000) settles at the lowest TOA plateau the campaign kept, $+0.70\,\mathrm{W\,m^{-2}}$. The costs were stated at adoption: NH March ice $+1.3\,\mathrm{M\,km^2}$, global T2m +0.21 K, and an AMOC that GM 1500 strengthens and GM 2500 gives back. **Drake Passage was not among the costs scored** (§8).

4.16 Rounds 37–40: the production line (2026–09–13→25)

Round 37 spent five branches on the Southern Ocean winter bias — salt plumes, Fox-Kemper MLE, a deeper lead-closing floor, all three together, and a calibrated basal melt — and **excluded all of them**. It is a freshwater regime, not a mixing deficiency. Round 38 moved the v3.5 defaults into `awiesm3.yaml`, started the line, and computed the sea-ice albedos from the model's own melt seasons rather than guessing: the four `ice_therm` albedos are *pond-free*, so darkening them to stand in for ponds double-counts, and the two snow albedos act in different hemispheres by measured surface state ($\partial\alpha/\partial$ `albsn` 0.908 in SH DJF against 0.149 in NH JJA). Round 39 scored four vertical mixing schemes and adopted `cvmix_TKE+cvmix_IDEMIX`, finding two real FESOM bugs on the way. Round 40 carried the line to 2100 and measured what it had become.

4.17 Rounds 41–46: albedo, the Labrador Sea, a pre-industrial reference, and the line as one record (2026–09–26→10–03)

Round 41 moved to albedo and started **PICAL_crunveg**, LPJ-GUESS restarted from the CRUNCEP state: it keeps 81 % of its larch after 19 years (pre-merge runs 34 and 20 %), and the forest costs about $2 \times 10^3\,\mathrm{km^3}$ of NH ice. Round 42 chased a lost Labrador Sea. The 1800 s deficit in IDEMIX mixing is a real numerical effect — the element–node–element round trip smooths once per step, so its daily strength scales as $1/\Delta t$ — and its fix (FESOM PR #1103) restored Kv but *not* convection; a noise-floor analysis then showed the build and step attributions rested on one run. A 61-run history search found the capped regime **only under Round 36's GM 2500 / Redi 1000**. Round 43 found the GM benefit is Southern Ocean ventilation, that neither the v3.3 ramp nor the Rossby radius can separate the Labrador Sea from the Southern Ocean margin on CORE3, and scored four 20-year arms: **hemispheric GM adopted**. Round 44 evaluated it (CMPI 0.861), found and fixed a factor-1000 snow-enthalpy error in `reval`'s Gregory plot, and re-scored the temperature against a PI reference: +0.46 K *warm*. Round 45 took the Antarctic pack apart: diffuse, not missing; a deficit that opens in autumn; drift that over-responds to correct winds. McPhee ocean-to-ice flux and `h0min` were ruled out. Round 46 put the five experiments of the production line on one axis (1850–2139, with an AMOC series) and re-scored Round 39's four mixing arms on heat by depth: IDEMIX, not TKE, warms the mid-depth ocean. It also took the tuning to the carbon-cycle configuration, at 6.7 times the cost.

4.18 Rounds 47–52: TKE alone, the v3.5.0 tags, Bug XIII, and the Southern Ocean cap (2026–10–03→06)

Round 47 ran TKE without IDEMIX: the mid-depth warming stopped (700–1500 m +42 → +9 mK/decade) and the surface warmed until `albsn` 0.82 held it flat at a PI bias of +0.55 K, but the Antarctic summer pack eroded ($-0.33 \times 10^6\,\mathrm{km^2}$ of February extent per decade) where an IDEMIX control arm from the same restart held it. IDEMIX keeps the Antarctic by burying heat, not by balancing the budget. Round 48 put every component of v3.5.0 on a tag and validated the tagged build against the line. Round 49 found **Bug XIII**: REcoM coupled to OpenIFS exchanged gases against a sea-level pressure of zero, so surface pCO_2 fell from 253 to 151 μatm in two years; fixed by coupling the OpenIFS pressure to FESOM (FESOM #1115). Round 50 returned to KPP with a uniform GM of 1000: the abyss warmed at the KPP-era +5 mK/decade, but the Southern Ocean convected over

twice the area and lost 40 % of its winter ice in eight years. Round 51 took that apart. The column is held by salt alone with a margin of 0.25 kg m^{-3}; the cap is 78 % of PHC3 in the open ocean and 31 % in the Weddell region in every run; `K_GM_max` 2500 is an effective 980 at 55–65 S, half the CORE2 value; the wind stress is only 10 % high; and two thirds of the Antarctic calving enters north of 60 S. Round 52 made that the working hypothesis.

4.19 Rounds 53–58: the last arms, the noise, and the CMIP7 spin-up (2026–10–06→15)

Round 53 attacked the cap through freshwater and ice instead of GM, which the AMOC (13–14 Sv) could not afford: four one-change arms from 2169. **Icebergs from the model's own budget** (calving minus basal melt, about 1100 Gt yr^{-1}, mapper calving off) work with no change to the `esm_tools` core and keep more Antarctic summer ice in all four decades, but the population takes decades to reach its melt rate. A higher **ice–ocean drag** doubles Antarctic summer volume and is a compensation (five times the measured value); **wave-model roughness** and **ice strength** do nothing. None strengthens the cap. Round 54 repeated the control with no change: it differs by 2.7×10^3 km^3 of Arctic September ice and 0.35 K at 60–90 N, passing the two-standard-error test on eleven of fifteen quantities, so the Arctic growth in three arms is not attributable. Round 55 lowered the **air–ice roughness** into the observed range: the 10 m wind over the pack sped up by 7–11 % and gave the stress back; the code is on the OpenIFS tag with the released defaults. Round 56 located the **abyssal drift** in the deep Atlantic and Southern Ocean, unchanged in rate for 360 years except when the mixing scheme changed; since no run length ends it, Jan decided on a fresh start from PHC3, run only until the carbon drift meets its threshold. Round 57 fixed the configuration and moved the tags (FESOM 2.8.2, OpenIFS 48r1v6). Round 58 is the spin-up: an iceberg pinned to one 3 km^2 cell crashed it in 1902 (fixed with slow drift for grounded bergs and a cap on iceberg cooling), and by 2009 it is colder than the tuning line, drifts at half its deep rate, and loses 0.41 GtC yr^{-1}.

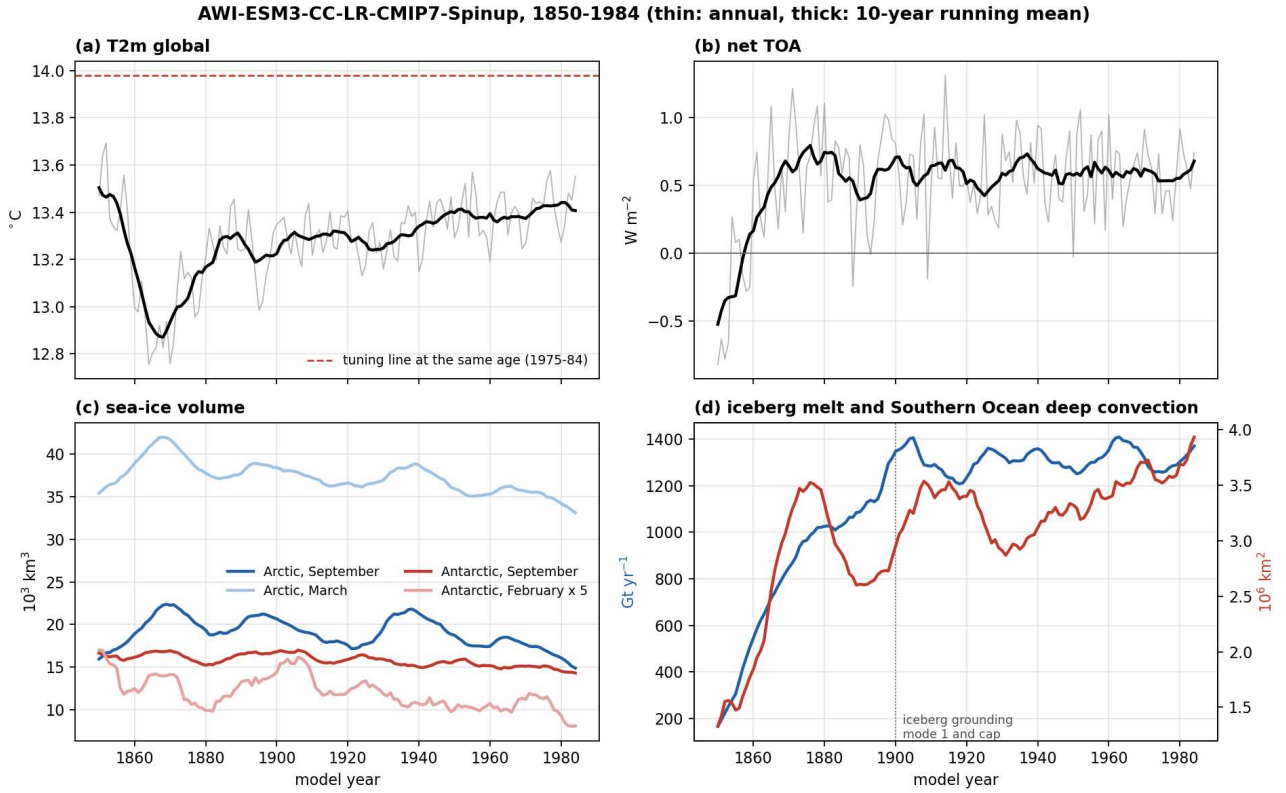


Figure 1: The CMIP7 spin-up, 1850–1984. (a) T2m recovers only to its starting value, 0.5 K below the tuning line at the same age. (b) The imbalance settles at +0.5–0.7 W m^{-2}. (c) Sea-ice volume. (d) Iceberg melt levels at about 1300 Gt yr^{-1} after the 1900 restart while Southern Ocean deep convection keeps spreading.

5 The production line, and the cold drift

PICAL (1850–1929) → **PICAL_momixoff** (branched 1920, to 2049) → **PICAL_ccnice** (branched 1940, to 2100). Throughput on levante, 47 nodes, 10-year legs: about 77 SYPD under TKE+IDEMIX, 86 under

`cvmix_TKE` alone, 75 under KPP, with ±15 % decade-to-decade scatter on identical configuration.

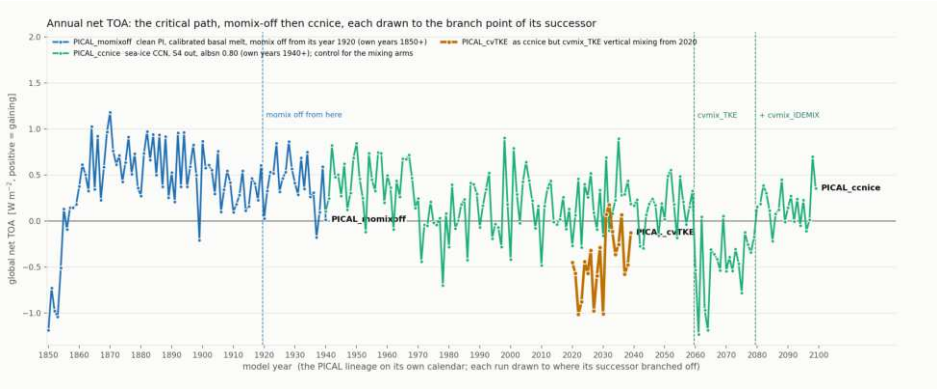


Figure 2: Annual net TOA along the critical path, each run drawn to the branch point of its successor. The two vertical marks are the mixing-scheme changes. `cvmix_TKE` at 2060 drives TOA negative; adding IDEMIX at 2080 brings it back. The run is never far from zero and never settles on it.

metric	2060–64	2095–99	trend/decade	95 % CI
T2m global	14.439	13.645	−0.214	0.025
T2m 60–90 N	−8.143	−9.842	−0.511	0.118
T2m 60–90 S	−15.650	−16.875	−0.238	0.104
NH extent, March	16.262	18.188	+0.522	0.103
net TOA	−0.775	+0.236	+0.272	0.079

Round 30 showed that tuning net TOA to zero *does* stop the ocean drift, so a run with positive TOA and a cooling surface needs explaining. It resolves in the ocean: total heat content is flat at +0.04 ZJ/decade over 2060–2099, while the bands are not (−17.8, −34.4, +19.4, +32.7 ZJ/decade for 0–100, 100–700, 700–2000 and >2000 m). **The upper 700 m is draining into the deep ocean at 0.32 W m^{-2} equivalent.** Split by mixing era, the mechanism tracks the scheme:

window	upper 0–700	deep >700	total	net TOA	net surface
KPP 2040–2059	+0.061	+0.164	+0.225	+0.136	+0.318
cvmix_TKE 2060–2079	−0.488	+0.176	−0.312	−0.468	−0.251
TKE+IDEMIX 2080–2099	−0.201	+0.452	+0.251	+0.167	+0.357

All in W m^{-2} equivalent; the surface budget includes the snow-enthalpy term, about 0.9 W m^{-2} globally, without which the budget is biased warm. `cvmix_TKE` alone flipped the upper ocean hard negative and the whole system lost energy. Adding IDEMIX halved the upper-ocean loss and drove deep uptake up, which is what internal-wave-driven diapycnal mixing should do. But the upper ocean is *still* losing heat, and that is what keeps the surface cooling. **This is the top open item.**

The crunveg line still cools, more slowly. Over crunveg 2101–2119 followed by ob1800 2120–2139, global T2m trends -0.059 ± 0.025 K/decade and 60–90 N -0.141 ± 0.115 (2σ); with gmhemi1800 as the continuation, -0.053 ± 0.025 and -0.203 ± 0.099. Over 2130–2139 gmhemi1800 has net TOA 0.52, net surface 0.67 and ocean uptake 0.69 W m^{-2}, of which 0.50 goes below 700 m.

The whole line on one axis. Stitched from PICAL, momixoff, ccnice, crunveg and gmhemi1800, the record runs 1850–2139. The warm layer at 300–1000 m builds through all of it; 700–1500 m is flat under `cvmix_TKE` alone ($+0.3 \pm 1.3$ mK/decade over 2061–2079) and warms from the leg IDEMIX came in (+27, then +28 on crunveg and +42 on gmhemi1800). The AMOC at 26.5 N spins up to 15.8 Sv by 1900 and holds 15–16.5 Sv through every atmospheric and sea-ice change until the 2120 branch, where it falls to 13.9 Sv: half of that is in ob1800 too, the rest is a northward shift under hemispheric GM with the 20–60 N maximum unchanged at 16.3 Sv.

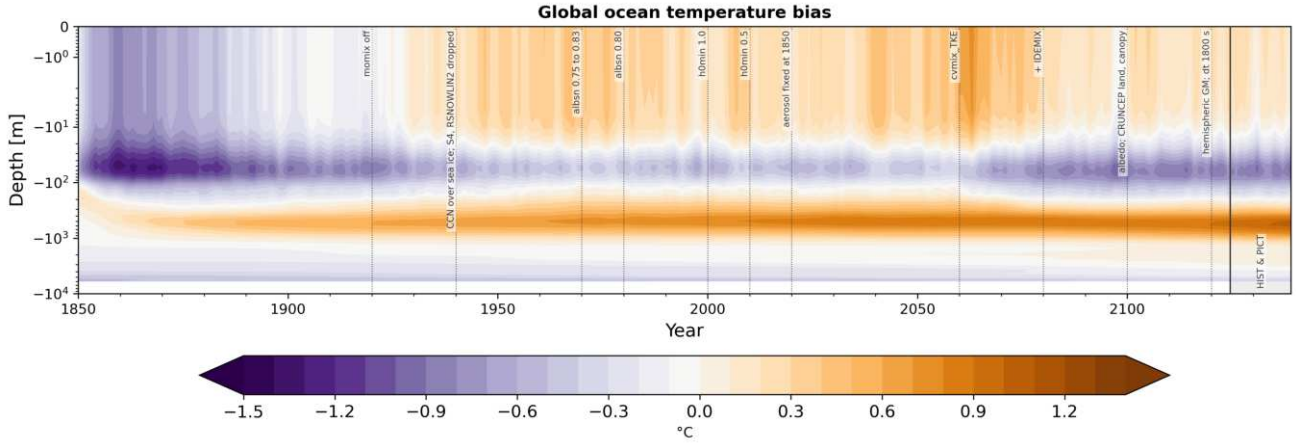


Figure 3: Global-mean ocean temperature against PHC3 along the production line, 1850–2139, with the eleven changes marked. The mid-depth warm layer builds throughout and no change along the line reverses it; the cold layer at 50–100 m weakens under KPP and returns with `cvmix_TKE`.

6 Vertical mixing: the four-way comparison

Four schemes, 20 years each (2020–2039) from one restart, differing only in `mix_scheme`.

arm	scheme	SH Feb vol	SH Feb ext	cavity melt
PICAL_ccnice	KPP (control)	—	—	1255.6
PICAL_cvKPP	cvmix_KPP	—	—	1283.5
PICAL_cvTKE	cvmix_TKE	1.31	1.95	1174.5
PICAL_cvTKEIDEMIX	cvmix_TKE+cvmix_IDEMIX	1.62	2.26	1062.1

Observed SH Feb volume about 2.07, extent about 3.0. Both KPP variants lose Antarctic summer ice, both TKE variants grow it, and `cvmix_KPP` reproduces the KPP control closely enough to be the null for sea ice (not for heat, below). **Adopted**, with the cost stated at adoption: NH April volume 38.04 rising +2.46/decade in the test arm, where TKE alone was flat at 35.17. On the production line the realised rate is $+1.145 \pm 0.968$, about half.

What Round 39 did not score: heat by depth. Global-mean temperature trends of the same four arms, 2021–2039, mK/decade:

arm	0–100 m	100–300 m	700–1500 m	1500–2500 m
KPP (control)	+28	+38	$+11.3 \pm 1.7$	−2
`cvmix_KPP`	−91	−52	$+15.0 \pm 0.9$	+3
`cvmix_TKE`	−214	−148	$+8.9 \pm 1.8$	+4
`cvmix_TKE+IDEMIX`	−307	−128	$+31.4 \pm 1.1$	+9

IDEMIX is what warms 700–1500 m. Both TKE arms drain the upper ocean, 0.26–0.34 K colder than KPP in the top 300 m after a decade, which is the heat drain of §5 at its source. KPP was the best of the four on this measure and the worst on Antarctic summer ice; it also kept the heat at the surface. And `cvmix_KPP` is 0.17 K colder than native KPP in the top 100 m, unexplained. **PICAL_crunveg_gmhemi_tke** (TKE without IDEMIX, 2140–2159 from gmhemi1800) is running.

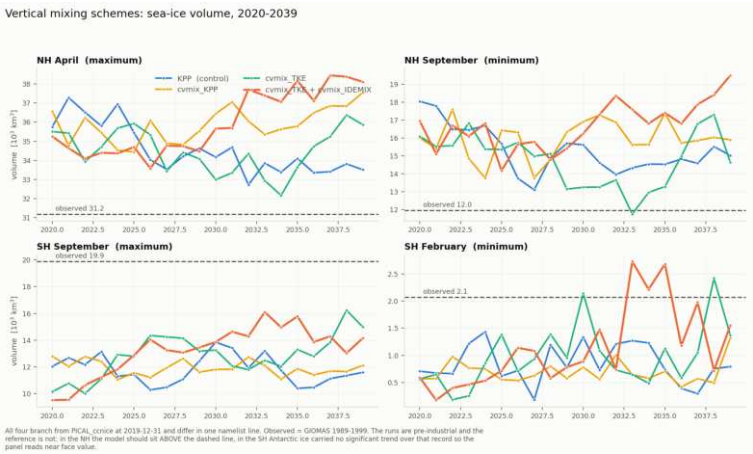


Figure 4: Hemispheric sea-ice volume across the four mixing arms, 2020–2039 from one restart. Both KPP variants lose Antarctic summer ice where both TKE variants grow it, and `cvmix_KPP` tracks the KPP control closely enough to serve as the null that shows the comparison is measuring the scheme rather than the branch.

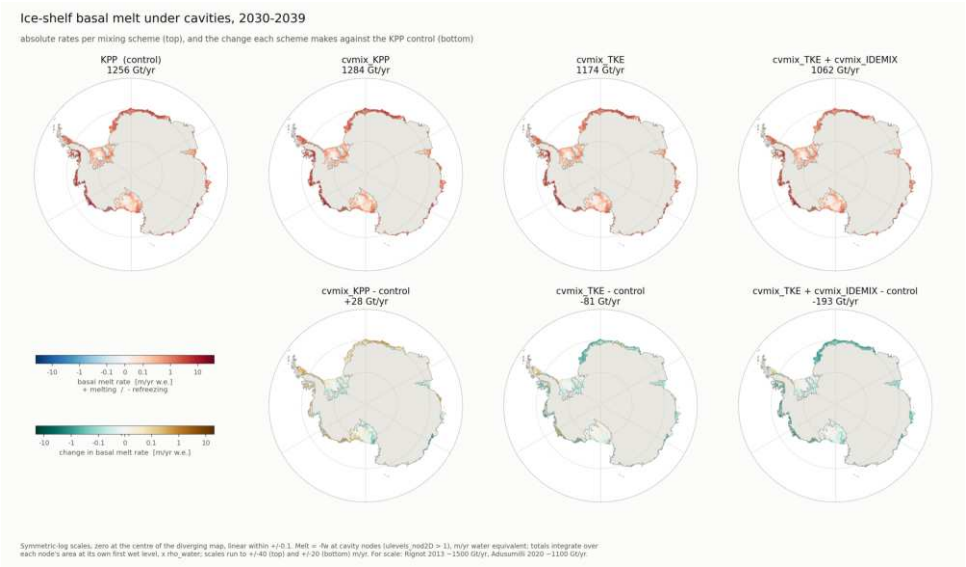


Figure 5: Ice-shelf basal melt under the cavities, 2030–2039, per mixing scheme (top) and the change each makes against the KPP control (bottom). Both TKE arms cool the cavities almost everywhere around the coast, with the largest reductions on the East Antarctic fringe and in the Amundsen/Bellingshausen sector; Ross and Filchner-Ronne interiors barely move. Symmetric-log scale, so refreezing keeps its own sign.

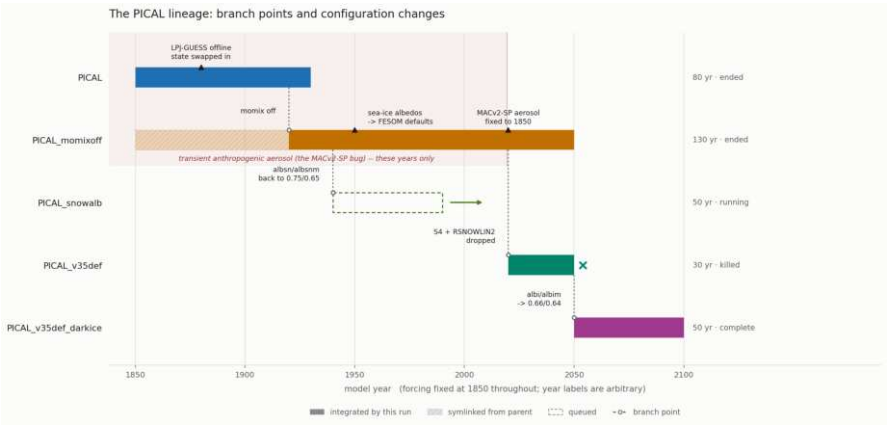


Figure 6: The PICAL lineage and every branch taken off it, on the model calendar. The production line is the spine; the arms hang off it at the year each was branched. The arms are cheap and the spine is expensive, which is the intended shape — but the spine has now run 160 years without converging.

7 The Southern Ocean now

Five more levers, all excluded (round 37). Four 20-year branches of **PI200** at 1560 against the PI200 control on identical years — **spp** (salt plumes), **mle** (Fox-Kemper), **h0** (lead-closing floor 0.5 → 0.25 m), **all3** — plus **cavg06** attacking the same loop from the ice shelves. None moves the winter over-ice bias in 20 years. The model sits in one of two freshwater regimes, salty-and-venting or fresh-with-warm-CDW, and these levers do not cross between them. The basal-melt calibration is worth keeping on its own terms (1-year screens gave 1038 / 1190 / 1402 $\mathrm{Gt\,yr^{-1}}$ at 0.5 / 0.6 / 0.7 against 1893 uncalibrated) and is upstream as FESOM PR #1064, but it is not a Southern Ocean fix. **SSS restoring stays excluded**; `surf_relax_s` is 0 and remains 0, even for Weddell convection.

The cloud shortwave deficit remains the other half. Across the whole coupled campaign no lever moved it until the mixing schemes: SW CRE at 45–65 S is −64.44 under KPP, −65.73 under `cvmix_TKE`, against CERES $-68.02\,\mathrm{W\,m^{-2}}$. `cvmix_TKE` closes about a third of the gap, more than any atmospheric lever achieved, and does it from the ocean side. That is worth recording: it is the one result suggesting the cloud error is not purely atmospheric — against round 30's finding that 84 % of it is already present in AMIP.

8 Drake Passage, and the price of the GM tuning

Computed with tripyview's model-accurate transect path, which resolves the 105 mesh edges the 66 W line crosses and integrates the flux FESOM's own discretisation carries.

run (1580–1599, identical restarts)	GM	resolved	**net**
PI200	resolution scaling (before)	110.0	**110.0**
PI200_gmR1500	Rossby cutoff, K_{GM}^{max} 1500	103.2	**103.2**
PI200_gmR2500	Rossby cutoff, K_{GM}^{max} 2500	93.3	**93.3**

16.7 Sv, 15 %, monotonic in K_{GM}^{max}, far outside the ∼1.5 Sv decadal scatter. The bolus term responds cleanly (±1.8 → ±2.7 → ±3.6 Sv) but its *net* is zero in every arm: the eddy-induced circulation across the passage is a closed overturning cell. On the production line the transport declines slowly, 105.1 (2040–49) to 98.9 (2090–99), with no step at either mixing-scheme switch, so the ACC weakness is structural rather than a cvmix artefact.

GM 2500 was adopted in round 36 for the mid-depth heat uptake, with the NH ice and AMOC costs priced. Drake Passage was not scored at the time. Both constraints are now on the ledger and they point opposite ways.

9 The Labrador Sea and the GM setting

What capped the Labrador Sea. Across 61 coupled runs and 468 run-decades, controlled for global SST, the persistent ice-capped Labrador state exists only under `K_GM_max` 2500 / `Redi_Kmax` 1000 — the Round 36 setting every PICAL run carries. GM 1000 runs stay at 0.05–0.3 March ice at every global SST from 17 to 20 °C; the clean pair PI200_gmR2500 / gmR1500 gives 0.684 against 0.280 on identical years (HadISST2 0.16). The time-step and build attributions made first rested on one run (ob1200) and are withdrawn; the IDEMIX per-step smoothing they pointed at is real, fixed in FESOM PR #1103, and does not restore convection.

Why CORE2 never did. AWI-CM3 v3.3 switched GM off below 30 km and its CORE2 Southern Ocean was coarse, so the Labrador Sea got about $113\,\mathrm{m^2\,s^{-1}}$ and 55–65 S about 2000. CORE3 refined the Southern Ocean margin to the Labrador Sea's spacing; production GM gives 983 and 981. Neither a resolution ramp nor the Rossby radius (8.0 km against 9.3 km) separates them.

Where the benefit comes from. In three GM 1500/2500 pairs the extra GM-1500 uptake enters through the Southern Ocean surface at 34–55 S (+0.19 to $+0.26\,\mathrm{W\,m^{-2}}$) and is stored in the Indo-Pacific at 700–2000 m: the AAIW/SAMW route. The subpolar North Atlantic releases heat. The Labrador capping is a separate North Atlantic side effect of the same parameter.

Four arms at 1800 s, 2120–2139, one restart, differences against ob1800 (GM 2500), "∗" beyond the paired $2\sigma/\sqrt{N}$:

2121–2139	gmferr1800	**gmhemi1800**	gmrd1800
T2m global (K)	−0.447*	+0.002	+0.105*
net TOA ($W\,m^{-2}$)	+0.372*	+0.032	−0.034
heat 700–2000 m ($W\,m^{-2}$)	+0.380*	+0.055*	−0.088*
NH March volume ($10^3\,km^3$)	+2.459*	+0.000	+0.150
SH September volume ($10^3\,km^3$)	−0.023	−0.066	−1.669*
NH March extent ($M\,km^2$)	+0.325*	−0.657*	+0.300
SH September extent ($M\,km^2$)	+1.380*	+0.114	−0.725*
Labrador March MLD, 2130–39 (m)	1723	2026	2487
Labrador March ice, 2130–39	0.18	0.10	0.07

ob1800 itself capped in its second decade (111 m, ice 0.43). **gmhemi1800 is adopted**: Labrador convection near 2050 m with no collapses and nothing else significant except NH March extent; it overshoots (ice 0.10–0.13 against 0.16), so NH 1500 is the next test. **gmferr1800** (Ferreira vertical scaling) is rejected for net TOA +0.46* in its second decade with the heat carried deep and the planet 0.5 K colder. **gmrd1800** (Rossby-radius K) is rejected for its Antarctic ice loss. The hemispheric gate breaks the campaign rule that regional differences must come from physics; it was accepted as an explicit exception after consulting the FESOM core developers.

10 The bias against a pre-industrial reference

ERA5 1990–2014 is not a PI target. Adding the HadCRUT5 warming from 1850–1900 to 1990–2014 per 5° box to (model − ERA5) gives a PI bias. Globally the model is −0.36 K against ERA5 and HadCRUT5 warmed 0.817 K, so **gmhemi1800 is +0.46 K too warm for the pre-industrial**.

PI bias (K), gmhemi1800 2125–2139	land	ocean	
JJA 30–45 N	+2.21	+1.42	NH summer mid-latitudes
JJA 45–60 N	+2.08	+1.37	
DJF 60–30 S	+1.94	+1.38	SH summer, the SO cloud deficit
DJF 60–75 N	−0.81	+0.43	Arctic winter cold
DJF 75–90 N	−2.08	−0.61	
ANN 30 S–30 N	−0.05	+0.05	tropics right

The maps add an Antarctic winter warm ring at 60–70 S, warm eastern-boundary upwelling coasts and a cold blob south of Greenland. Caveats: HadCRUT5 1850–1900 coverage is thin at the poles, and the AMIP pair showed the model under-warms Siberia between epochs, so an observation-based PI target is not the model's own PI. **This reframes the founding trade-off**: against ERA5 the model was too cold with a positive imbalance; against a PI reference it is too warm with a positive imbalance, and the two point the same way.

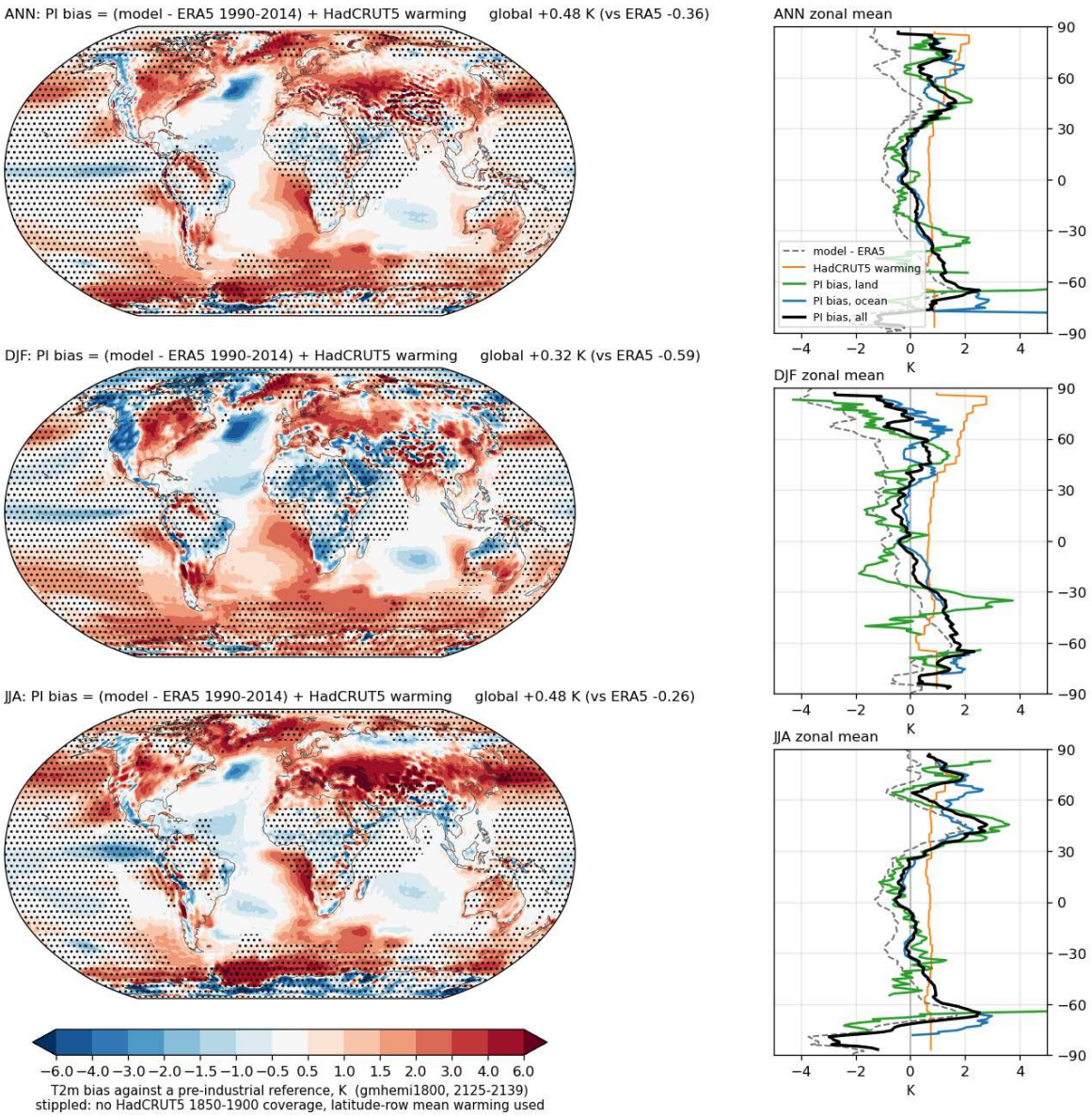


Figure 7: T2m bias of gmhemi1800 2125–2139 against a PI reference, (model − ERA5 1990–2014) + HadCRUT5 warming 1850–1900 → 1990–2014. Stippled: no HadCRUT5 1850–1900 coverage.

The warm NH summer land is dry and sunny. Central Asia 40–55 N in JJA is +2.9 K against ERA5 with 0.7 against GPCP's 1.3 mm d^{-1}, +20 W m^{-2} surface SW against CERES, an evaporative fraction of 0.25 and top-layer soil water 0.09; Europe is +1.3 K, N America +1.6, Siberia −1.1. LPJ-GUESS cover is used directly by OpenIFS when coupled (no double counting), and it is sparser than the IFS default: effective JJA cover 0.31 against 0.57 over central Asia, 0.69 against 0.87 over Europe, 0.65 against 0.79 over N America. Whether the sparse vegetation causes the dry summer or follows from it cannot be separated from this run.

11 Sea ice: volume is the state variable

Against GIOMAS 1989–99 the NH sits above observations in every month, which a pre-industrial run should, but by more than the epoch offset justifies in April; the SH is the observed cycle scaled down by about a third in every month, and that comparison is close to like-for-like because GIOMAS has no significant Antarctic trend across the window. Extent, scored against OSI-SAF, is reported beside volume throughout — with the caveat that the record on disk has no December, 286 months rather than 312, and a December value must not be synthesised. Extent is geometrically saturated in this configuration and has hidden a real Arctic runaway before: across the 1950 albedo step it barely moved while volume climbed from 24 to 48×10^3 km^3. **Volume is the state variable that says where the pack is going**; extent is the one that gets compared to satellites. An ice-affecting lever is scored on volume first, and a null result in extent is not evidence that nothing happened.

The Antarctic pack is diffuse, not missing (round 45). In SH JJA the area where both model and HadISST have ice is 37.7 % of the domain and +1.77 K against ERA5, at concentration 0.74 against 0.84; ice the model misses is 2.4 %. The 50–90 S area deficit opens in April–May (−2.6 to −2.9 M km^2) and stays flat through September while the dynamic loss peaks, consistent with round 33's autumn entrainment of the +0.9 K layer at 100 m. Against OSI-455 drift 2005–2014 the SH ice moves 1.29 times too fast; Antarctic-wide deformation is about right (ratio 0.95) but wrong by sector, with Ross and Amundsen–Bellingshausen diverging at +1.68 and +1.26 % d^{-1} against +0.26 and −0.07. The winds are not the cause (7.2 against NCEP-2's 8.0 m s^{-1}, the Amundsen Sea Low at the same relative depth): the ice over-responds, wind factor 2.4 against 1.4 %, and turns +63° in the Ross against +24° observed. Slowing it would tighten that pack but likely cost SH extent and add Arctic volume, so it is documented as a known bias, not tuned. A McPhee St^*u^* ocean-to-ice heat flux went the wrong way in both hemispheres (SH September −1.25*, NH +1.04* × 10^3 km^3), and `h0min` is hard-coded in the coupled build.

12 The boreal forest, and what it blocks

Siberian box 55–75 N / 60–180 E, year 2099, 684 cells:

state	BNS	TREEFPC	GRASSFPC	grass > tree
offline CRUNCEP (the restart)	0.227	0.415	—	29.2 %
11E after 50 coupled years	0.017	0.083	—	69.6 %
PICAL_ccnice 2099	**0.001**	**0.050**	0.216	**84.9 %**

Stable between 2090 and 2099, so this is the settled state, and it is the end point of the transient round 27 identified: the coupled model dismantles an inherited forest, 92 % of the larch in 50 years. **The mechanism is competition with C3 grass, not a climate gate**: BNS has no cold limit at all, the only temperature gate is a warm one 34 K away, and GDD5 clears on 91 % of cells in the very run where the larch dies. Questions about the boreal forest should be answered through the grass-versus-larch competition, not through thresholds.

This is now a practical blocker. Laszlo Hajdu's LPJ-GUESS canopy work (stem area index via `ltos`, shrubs routed to low IFS vegetation, `iftreefracca 1`) measured +0.46 K global T2m in a run that *has* a boreal forest, and improved its ERA5 comparison (bias −0.46 against −0.92 K; RMSD 1.38 against 1.77). It acts on the tree side of the tile. With 5 % tree cover here, that result is not transferable, and a continuation would measure what the change does to a grassland. **PICAL_crunveg** — identical except that LPJ-GUESS starts from the CRUNCEP spin-up end state — was built to answer this, ~~and cancelled before starting when the campaign moved to albedo~~, and ran on albedo from 2100.

Round 41: the forest survives on the merged code, and costs ice. ccnice does not regrow (Siberian tree FPC 0.062–0.067 per decade after 2100). crunveg keeps 81 % of its BNS after 19 years against 34 % (080a) and 20 % (11G) from the same CRUNCEP start, also at matched GDD5 and grass PAR — not yet separated from state migration or non-GDD climate. Paired against ccnice over 2101–2119, the forest gives global T2m +0.24 K, 60–90 N +0.91 K and NH April / September volume −2.0 / −2.35 × 10^3 km^3, putting the NH at or below GIOMAS 1989–99.

13 Model skill: the degradation is tropical

CMPI, lower is better, across four windows of the same run:

	1980–89	2000–09	2080–89	2090–99
CMPI global	0.844	0.879	0.902	0.903
nino34	0.966	1.064	1.124	1.121
tropics	0.877	0.874	1.013	1.011
northmid	0.823	0.834	0.867	0.869
arctic	0.867	0.888	0.880	0.871
antarctic	0.718	0.790	0.717	0.726

The polar regions are flat (+0.003 arctic, +0.008 antarctic across 110 years). The tropics carry the whole degradation (+0.155 nino34, +0.134 tropics). By variable the worst movers are `tas` +0.149, `thetao` +0.127, `zg` +0.111. **No mechanism is attached to this yet**, and it is not where the campaign has been looking. For scale, the campaign's best-scoring run was 11R at 0.7577.

On the crunveg line with hemispheric GM (gmhemi1800 2125–2139, round 44): CMPI **0.861** against crunveg 2101–2119 0.870 and ccnice 2101–2129 0.865, with 296 of 400 entries better than CMIP6 against 286. Gains: Niño3.4 SST and T2m, NH mid-latitude sea ice, tropical 300 hPa wind, SH mid-latitude MLD, Antarctic ice. Losses: NH mid-latitude MLD (the Labrador overshoot), SH mid-latitude 10–100 m salinity, Arctic surface salinity and winds, NH mid-latitude 1000 m temperature.

14 Levers, across the whole campaign

lever	verdict	the number
G4 (`RVRSMIN` cap)	adopted	Siberian JJA +0.952 K, $t = 7.68$; SO SW RMSE 6.88 → 4.80.
K1 (snow-free albedo)	adopted	+1.036 K, RMSE 1.579 → 1.524, no Arctic penalty.
P3/P5 (`ECE_SNOW_SCF` 3)	adopted	Refitted to snow courses; beats scheme-off on bias, RMSE and JJA against 105 RIHMI stations.
`RSBLB` 2.0	adopted	Arctic DJF screen +0.614 K coupled, net TOA null. The first lever to move winter and summer the same way.
GM 2500 / Redi 1000	adopted	Lowest TOA plateau kept, +0.70 W m^{-2}. Costs NH ice, AMOC, and 16.7 Sv of Drake.
`cvmix_TKE+IDEMIX`	adopted, under review	Closest to the observed Antarctic summer pack; costs NH April volume, and IDEMIX warms 700–1500 m at +31 mK/decade against +9 for TKE alone. A TKE-only arm is running.
`cavity_gamma_scale` 0.6	adopted	1190 Gt yr^{-1}; upstream, FESOM #1064.
hemispheric GM (2500 S / 1000 N)	adopted	Labrador MLD ∼2050 m, no collapses; T2m, TOA, Siberia, ice volume null; NH March extent −0.66*. Overshoots: ice 0.10–0.13 vs 0.16. FESOM #1112 (open).
CRUNCEP land state (crunveg)	adopted	BNS 81 % retained after 19 years; -2×10^3 km^3 NH ice, +0.24 K.
Ferreira GM scaling	rejected	Most realistic Labrador ice (0.17–0.18), but TOA +0.46* in decade 2, heat carried deep, global −0.5 K.
Rossby-radius GM	rejected	SH September volume $-1.67^* \times 10^3$ km^3, extent −0.73*.
GM 1500; v3.3 GM ramp	not adopted	Both open the Labrador Sea and give back the uptake benefit (TOA +0.38*, +0.44*); the ramp loses 40 % of SH September volume.
IDEMIX smoothing `dtref`	fix, not a lever	Removes ∼85 % of the 1800 s mixing deficit; Labrador 414 vs 370 m, no convection. FESOM #1103 (open).
McPhee St^*u^* ice–ocean flux	rejected	SH September −1.25*, NH $+1.04^* \times 10^3$ km^3: wrong way in both hemispheres.
`h0min`, `h0/h0_s`	not a lever	Hard-coded in the coupled build; the namelist values are inert.
slowing the Antarctic ice drift	not tuned	Winds right, ice over-responds; would likely cost SH extent and add Arctic volume.
S4 `RCL_INPPMIN` 50000	**retired**	Headline of the 2026–08–11 summary. Siberian JJA +1.194 K AMIP, −0.365 K coupled.
I3	rejected	Strongest summer lever built (+1.322 K) at +1.72 W m^{-2} at 60–90 N.
`tanh` snow depletion	falsified	−20.2 K against 174 676 station soil observations; unrepresentable complete cover.
`LAMSK`, moss proxy, D1	falsified	Skin conductivity +0.125 K against a 0.41 threshold; the moss proxy moved monotonically the wrong way; `RCAPDCYCL` 4 null.
Raupach z_0	dead structurally	Needs a canopy; makes the winter surface 4.4× smoother as the model stands.
DMS / marine CCN	priced out	Best SO lever found (−7.25 W m^{-2} DJF) but the tropical tolerance binds below the published coefficient floor. Reconsidered as a global cooling lever in round 44 and declined for the last round.
`RSNOWLIN2` 0.04	not adopted	Only lever that moves the tropical longwave (+2.74); costs +0.42 globally.
SB3 mixing length	falsified	3.43 tropical SW per K against `RSBLB`'s 0.88.
salt plumes, MLE, `h0min` ↓, all three	excluded	Round 37, no SH winter response in 20 years.
`h0min` 0.5 → 1.0	reverted	Flux response as designed (1.71→1.10), state did not follow; 60–90 S bias +0.82 → +1.45 K.
darkening `albpnd`	falsified	Wrong direction, and double-counts the ponds.
SSS restoring	excluded by policy	Permanently.

15 Reversals on the record

Kept because a summary that only lists what survived misrepresents how the campaign learned.

- ~~S4 is the best configuration the campaign has.~~ Its benefit reverses sign coupled. The 2026–08–11 summary led with it.
- ~~The period offset is ≈1.12 K, so the boreal bias is ≈ −1.0 K.~~ Measured directly at +0.42 K; the bias is ≈ −2.0. The method error was applying an *observed* epoch change to a model that does not reproduce it.
- ~~There is a −1.0 W m^{-2} structural leak in the coupled column.~~ It was the omitted snow-enthalpy term.
- ~~The boreal problem is competition, not climate.~~ Both, and the competition is driven by the forcing: warming recovers 59 % of the gap at +2 K while `grass>tree` stays at 43–45 %.
- ~~Cloud amount is unreachable, so the SO needs an opacity lever.~~ Round 33 showed opacity is the right correction for 90–60 S and the *wrong* one for 60–45 S, which every campaign lever had been applying.
- ~~The ice is too dark.~~ It is too diffuse: compact-ice albedo is 0.760–0.778, but no cell exceeds 90 % concentration in January–March.
- ~~Positive net TOA means the run is warming.~~ It cools at 0.21 K/decade.
- ~~The 2120s Labrador collapse is the new build and the 1800 s IDEMIX step.~~ Every such difference was against one run (ob1200); the capped regime is GM 2500 / Redi 1000.
- ~~The GM 2500 uptake benefit survives switching GM off on the fine mesh.~~ The v3.3 ramp gave it all back and lost 40 % of SH September volume.
- ~~The model is too cold.~~ Against ERA5, yes (−0.36 K); against a PI reference it is +0.46 K warm.
- ~~OpenIFS double-counts LPJ-GUESS vegetation cover.~~ `RVCOV` is applied only when LPJ-GUESS is not coupled.
- ~~The steppe LAI of 4.9 is too high.~~ Per-tile LAI; cover-weighted 0.82.
- ~~The Antarctic winter warm ring is a missing ice edge.~~ It is a diffuse pack; the missing-ice class is 2.4 % of the area.
- ~~The surface imbalance is 1.51 W m^{-2}.~~ `reval` dropped the snow enthalpy; corrected 0.67.
- ~~`cvmix_KPP` is the null of the mixing comparison.~~ For sea ice; its top 100 m is 0.17 K colder than native KPP.
- ~~KPP with a uniform GM of 1000 gives a quiet deep ocean and a surface the atmosphere can correct.~~ The abyss was quiet; the Southern Ocean opened and net TOA went to +1.3.
- ~~The Southern Ocean wind stress is 35 % too strong.~~ That was u_{10}^2; the stress is +10 %.
- ~~Icebergs, wave roughness and ice–ocean drag grow Arctic ice.~~ An unchanged second control moved half as far the same way.
- ~~Lowering the air–ice roughness slows the Antarctic drift.~~ The 10 m wind over the pack speeds up and returns the stress.
- ~~Continue the 350-year state rather than restart.~~ Advised, then reversed the same evening: the drift rate is the same either way, but a restart begins the deep Atlantic 1.2 K closer to observations.
- ~~Spin-up land carbon is 2553 PgC.~~ The 1° LPJ-GUESS files count whole coast cells; rebuilt from the native land points it is 2260.

16 Bugs and traps that cost this campaign

what	the lesson
A dead namelist block	`NCMIPFIXYR` sat in `&NAMECECMIP6`, which the runtime does not read, so the AMIP runs were never at 1850 forcing. Krakatoa in 1883–86 is the fingerprint.
Two OASIS traps	`oasis_put` returns the waitgroup code for every field in a group but the last, so the ice-skin anchor was 0.0 in all 63 923 logged events and the skin reached −36 470 K. And `GSMART` on a sign-changing field gave 9105 excursions above 1 kW m^{-2} onto real ice in one coupled year, where `GSSMAR` and `GLOBAL` give none.
Implicit netCDF fill value	`read_other_NetCDF` left `miss` at 0.0 when a variable declared no `missing_value`, masking 146 694 real zeros and admitting 9.96921×10^{36}. FESOM PR #1087.
`MPI_DOUBLE_PRECISION` on a `real(WP)` buffer	Five `MPI_AllREDUCE` calls read 8 bytes from a 4-byte variable in a single-precision build. Not only a diagnostic: the corrupt total was fed back into the physics through `idemix_botforc_Etot`. Same PR.
The staged library is a symlink	`<exp>/bin/fesom/lib/fesom/libfesom.so` points into esm_tools' per-leg snapshots and `md5sum` follows it. The `fesom` executable is a driver stub, byte-identical across builds, so a matching binary checksum proves nothing. Use `ls -l` before `md5sum`. Cost a 47-node slot and a NaN'd ocean.
The instruction file nothing reads	`guess.ins` is hard-wired in `framework.cpp:1063` and imports `ecearth.ins`; `global.ins` and `run_coupled_4_1_2.ins` are staged and never reached. The two had drifted apart, `iftreefracca` 1 against 0. A parameter put in the wrong one is silently inert.
IDEMIX smooths once per step	The element–node–element round trip of the internal-wave energy is an un-Δt-scaled smoother, so 1800 s mixes less than 1200 s; the tuned state leans on it (Kv −12 to −19 % without it). FESOM PR #1103.
L_f in the wrong units	`reval`'s Gregory plot multiplied snowfall in metres by 334000 instead of $\rho_w L_f = 3.3355 \times 10^8$, dropping a 0.81–0.86 W m^{-2} term and reading 1.51 for 0.67. `release_evaluation_tool2` PR #7. The same term behind round 30's "structural leak".
Inert lead-closing namelist	`h0`/`h0_s` are read only by `ice_thermo_oce.F90`, compiled out when coupled; `ice_thermo_cpl.F90` hard-codes `h0min` 0.5. And the `/ice_therm/` read (`ice_init.F90:123`) never checks `iostat`, so one unknown name silently resets every later name in the group.
One realisation	A single 1200 s arm sitting at the 28th percentile of the null made the build and step look guilty for two days. Test against the null before attributing.
Chained legs and the date file	Legs read a per-leg archive, not the source runscript, so edits need `-U`; and the date file advances past a crashed leg, so it is not evidence of progress.
REcoM brings its own namelists	`bin_2p3z2d/namelist.io` is a different output list (37 outputs of the physics run gone), and the CMIP7 XIOS templates defined no BGC field, so PICAL_cc's first leg wrote no REcoM diagnostics. Both fixed in `esm_tools fdde4730e`. Diff the resolved namelists against the physics run before submitting; that diff also showed the suspected return of salinity restoring was not real.
Bug XIII, a pressure of zero	REcoM is built without `__oasis`, so it read FESOM's `press_air`, which a coupled run never fills: every coupled REcoM run exchanged gases against a vacuum. The giveaway was `dpCO2s` = `pCO2s` +6.7 in every month.
The significance test is too generous	Two standard errors of the interannual spread do not contain the decadal spread between realisations: an unchanged second control passed it on eleven of fifteen quantities.
A berg fifty times its cell	FESOM draws a large iceberg's melt heat from one mesh cell; with grounded bergs held still, one cooled a 3 km^2 cell to −5.6 °C and crashed the spin-up in 1902.
Remapped land carbon	A conservative remap of land-only values to 1° keeps the mean over the land part and loses its size: totals 12 % high. Rebuild the land fraction from the native points, and keep the native file (`-nc_keep_unstruc`).

17 Closed by measurement

Claims that were live at some point and are now settled, each with the observation that closed it.

claim	what closed it
The cold bias is a tuning artefact	Centred RMSD is 1.41–1.64 K across all 29 early runs, a ±8 % spread. The runs differ by an offset.
Land surface albedo explains the universal bias	Removing 74 % of a measured bias bought 0.089 K of ≈0.7 K.
Boreal needleleaf is excluded by a cold gate	True for BNE (0.7 % of cells clear it) and false for BNS, which is ≈80 % of NE Siberia and is outcompeted instead.
ERA5 is an adequate cloud reference for the Southern Ocean	ERA5 is IFS and reproduces the model's own bias; CERES and MODIS agree with each other against both.
The tropical deep warming is diapycnal flux	It is heave: 0.42 of the +0.477 K at fixed depth is the isopycnal sinking +42.4 m.
The SH winter bias is a vertical-mixing deficiency	Four levers, four nulls, 20 years each from one restart.
The ice is too dark because the ponds are too bright	The `ice_therm` albedos are pond-free; darkening them double-counts.
The albedo sensitivities are fixed properties	NH $\partial\alpha/\partial$ `albsn` moved 0.149 → 0.504 as the Arctic cooled.
Positive net TOA means the run is warming	It cools at 0.21 K/decade; the ocean redistributes rather than accumulates.
GM tuning is free at the Southern Ocean	16.7 Sv of Drake Passage transport.
GM tuning is free in the North Atlantic	The capped Labrador regime exists only under GM 2500 / Redi 1000 (61 runs; clean pair 0.684 against 0.280).
The Labrador collapse is an IDEMIX time-step effect	The smoothing fix removed 85 % of the Kv deficit; convection stayed at 414 against 370 m.
The GM uptake benefit is in the North Atlantic	Three pairs: it enters at 34–55 S and is stored in the Indo-Pacific at 700–2000 m.
The Rossby radius separates the Labrador Sea from the SO	8.0 against 9.3 km; gmrd1800 lost 1.67×10^3 km^3 of SH ice.
The model is too cold for a PI control	+0.46 K against an epoch-corrected reference; only the Arctic winter is cold.
The Antarctic drift error comes from the winds	Wind speed and the Amundsen Sea Low are close to NCEP-2; the ice over-responds (2.4 against 1.4 %).
Dropping IDEMIX costs nothing	The Antarctic summer pack erodes under TKE alone (-0.33×10^6 km^2 per decade) and holds with IDEMIX, same years and restart.
A GEOMETRIC closure would give the north–south GM contrast	Its growth rate is 1.4–1.6 times the dissipation in the north and 0.6–0.9 in the south.
Freshwater placement or sea-ice dynamics can restore the cap	Icebergs, wave roughness, ice–ocean drag, ice strength, air–ice roughness: 0.21–0.26 against 0.32.
A longer spin-up ends the abyssal drift	No curvature in 360 years; only the mixing scheme changed the rate.

18 Open, in priority order

Re-ranked 2026–10–15 after rounds 47–58.

1. **When the spin-up meets the carbon criterion.** Ocean and land together lose 0.41 GtC yr^{-1} over 2000–2009 (ocean 0.17, land 0.24) against 0.1; the land loss has moved to soil and litter, which respond slowly.
2. **Whether a spin-up 0.4–0.5 K colder than the tuned state is acceptable.** The forest lost in its first decades and the higher land albedo are the candidate; continuing will not close it.
3. **The abyssal drift**, +6 mK/decade in the spin-up and +12.5 late in the tuning line, in the deep Atlantic and Southern Ocean. Why every CVMix scheme warms the abyss twice as fast as FESOM's KPP is open; the piControl will carry the drift.
4. **The Southern Ocean cap at 70–80 % of observed**, with the levers that spare the AMOC exhausted. Documented as structural, with the Southern Ocean cloud deficit and the diffuse pack.
5. **Icebergs**: a large berg's melt is drawn from one cell, and the cooling cap that prevents a crash discards about 1 % of the melt heat. Changing leg lengths untested.
6. **The Arctic at `albsn` 0.82** may be only marginally stable: three of four arms gained 4–6.5×10^3 km^3, half of that within the noise of a second control.
7. **The warm, dry NH summer land**, the **ACC 30 % weak** and the **stratocumulus coasts**: structural, to be documented.

Superseded (2026–10–03 list): the Labrador overshoot under hemispheric GM (NH 1500 was not run; the release kept 1000) and the cooling of the line (the TKE-only line with `albsn` 0.82 is flat).

~~Does the canopy work do anything without trees? `PICAL_crunveg` was built to answer it and cancelled before starting.~~ It ran; the forest survives (round 41). ~~Will the CRUNCEP state load on `TC095-land`?~~ It does: crunveg ran from it.

19 What would change the picture

Stated in advance, so that a result which contradicts them is recognised as such.

- **If the upper-ocean heat drain stops without the surface warming**, then the cold drift is not driven by the redistribution and the diagnosis in §5 is wrong.
- **If NH April volume flattens on its own** over the next 50 years, the cost priced against TKE+IDEMIX at adoption was an artefact of the 20-year test arm and the scheme is cheaper than believed.
- **If a Southern Ocean lever moves the winter bias in under 20 years**, then the two-regime picture is wrong, or the regime boundary is closer than the five excluded levers suggest.
- **If Drake Passage recovers without GM being reduced**, the 16.7 Sv attribution is confounded with something else that changed between `PI200` and the arms.
- **If the canopy work produces +0.46 K here**, with 5 % tree cover, then it is not acting through the tree canopy and the mechanism is misattributed.
- **If the tropical CMPI degradation appears in a run without the cold drift**, the two are independent and the tropics need their own investigation.
- **If closing the remaining cold bias does not recover the forest**, the round-27 transfer function does not transfer coupled, and the campaign's founding premise fails at the last step.
- **If a second realisation of a GM 1000 NH arm caps the Labrador Sea**, or GM 2500 runs convect in a second realisation, the GM attribution of round 42 rests on noise as the build attribution did.
- **If the crunveg forest decays like 080a/11G on an offline old-binary rerun with the same forcing**, its survival is the climate, not the merged canopy code.
- **If the spin-up's carbon drift stalls above 0.1 GtC yr^{-1}**, the land soil pool is not converging on a spin-up time scale and the criterion, not the run, has to change.
- **If the spin-up warms back to the tuning line's 13.9 °C without the forest returning**, the forest is not what holds it cold.
- **If the abyssal rate in the spin-up rises to the tuning line's +12.5 mK/decade**, the lower rate so far is the young state, not the configuration.

20 The run inventory since the last summary

Every coupled arm that produced a scoreable result since 2026–08–24; the 2120 branches all start from crunveg's 2120 restart. All are 20 years unless stated, all branch from one restart with the control on identical years. The campaign's earlier arms — the 29 screening runs, the AMIP A/D/F/G/I/K/L/P/S series and the 11-series coupled roster — are catalogued in `report.tex`, appendix `app:params`.

run	years	the one thing it changed, and what it showed
PI200_spp / _mle / _h0 / _all3	1560–79	Salt plumes, Fox-Kemper MLE, lead closing `h0min` 0.5 → 0.25 m, and all three together. No SH winter response in any, and no combination effect.
PI200_cav05/06/07	1600	One-year basal-melt screens: 1038 / 1190 / 1402 Gt yr^{-1} against 1893 at 1.0. 0.6 taken forward.
PI200_cavg06	1600–19	`cavity_gamma_scale` 0.6. Melt calibrated; no Southern Ocean response.
PI200_gmR1500	1580–99	Rossby cutoff, K_{GM}^{max} 1500. Drake 103.2 Sv.
PI200_gmR2500	1580–99	Rossby cutoff, K_{GM}^{max} 2500. Drake 93.3 Sv; the adopted setting, and the 16.7 Sv cost.
PICAL_v35def	2020–49	`RCL_INPPMIN` 50000 vs 70000, paired with momix-off. Context for the TOA figure.

run	years	the one thing it changed, and what it showed
PICAL_v35def_darkice	2050–99	`albi`/`albim` darkened against its parent. An Arctic runaway fifty years did not arrest.
PICAL_cvKPP	2020–39	`cvmix_KPP`. The useful null: reproduces the KPP control.
PICAL_cvTKE	2020–39	`cvmix_TKE`. Grows the Antarctic summer pack; costs energy.
PICAL_cvTKEIDEMIX	2020–39	`cvmix_TKE+cvmix_IDEMIX`. Closest to the observed summer pack; adopted.
PICAL_ccnice	1940–2129	The production line to 2099; continued on albedo to 2129 as the unforested control.
PICAL_crunveg	2100–19	LPJ-GUESS from the CRUNCEP state with the merged canopy code. BNS 81 % retained; the new production line.
gm2500 (`_ob1200`), **ob1800**	2120–39	Old build at 1200 / 1800 s (ob1200 to 2129). The GM 2500 references.
ts1200	2120–29	New build at 1200 s. Labrador 520 m.
ihf0 / ihf1	2120–29	New build at 1800 s, McPhee flux off/on. Wrong way in both hemispheres.
nx1800	2120–29	ihf0 + IDEMIX smoothing fix. Kv restored, Labrador 414 m; second leg cancelled.
gm1500 / gmramp	2120–29	1200 s, GM 1500 / v3.3 ramp. Both open the Labrador Sea and give back the uptake benefit; second decades cancelled.
gmferr1800	2120–39	Ferreira scaling. Rejected, TOA $+0.46^{*}$.
gmhemi1800	2120–39	Hemispheric GM. Adopted.
gmrd1800	2120–39	Rossby-radius GM. Rejected, SH ice.
PICAL_cc	2140–44	gmhemi1800 tuning on `develop-cc`: REcoM cold start, concentration-driven PI CO_2. 0.674 s per model hour; no BGC diagnostics in this leg; paused.
gmhemi_tke	2140–59	gmhemi1800 with `cvmix_TKE`, IDEMIX off. Mid-depth warming stops, the surface warms.
tke_albsn082	2160–2209	`albsn` 0.82 on the TKE-only line. Surface flat, PI bias +0.55 K, Antarctic summer pack eroding after 2190; the control of every later arm.
idemix_albsn082	2170–2209	IDEMIX back on. Holds the Antarctic, buries heat at 700–1500 m (+29 mK/decade).
kpp_gm1000	2170–89	KPP and uniform GM 1000. Quiet abyss; Southern Ocean opens, net TOA +1.3; second leg not run.
PICAL_v350val / cc_v350val	2190–99, 2140–41	The tagged v3.5.0 build, physics-only and carbon-cycle. Within noise of the line.
icb	2170–2209	Icebergs from the mass budget. Works; Antarctic summer ice above the control in every decade; cap +0.012.
wamz0	2170–89	ECWAM default sea-surface roughness. No effect.
cdoi10	2170–89	Ice–ocean drag 0.010. Antarctic February volume doubled; a compensation; cap weaker.
cpres15	2170–89	`c_pressure` 15. No lasting effect.
ctl2	2170–89	The control repeated with no change: the noise yardstick.
z0def / z0floor / z0form	2170–89	Newer build, and the air–ice roughness halved in the floor, then also in the form drag. Build validated; roughness returned by the wind.
TEST_ccspin_1y, **SPIN_cc_test1**	1850, 1850–99	Smoke test and 50-year test spin-up on `develop-cc`: icebergs with REcoM on 2688 ranks from PHC3.
AWI-ESM3-CC-LR-CMIP7-Spinup	1850–	The CMIP7 spin-up on `v3.5.0-cc`, run by a colleague; at 2009 on 2026–10–13.

21 The evaluation protocol, and which references are independent

Every window is scored the same way, so results from different rounds are comparable.

target	reference, and its caveat
Sea-ice volume	**GIOMAS**. A model with assimilation, not a measurement: Arctic volume carries 10–20 % uncertainty and Antarctic more. **Use 1989–99** for a PI run — the Arctic thinned sharply across the full record, the Antarctic has no trend.
Sea-ice extent	**OSI-SAF**. The record on disk has **no December** — 286 months — and a December value must not be synthesised.
Radiation	**CERES**. Period offsets matter, and measuring one rather than estimating it once reversed a conclusion.

target	reference, and its caveat
Screen temperature	**ERA5**, except for anything the IFS land surface decides: ERA5 is an HTESSEL sibling and reproduces the model's own cloud and snow biases.
Soil temperature	**RIHMI-WDC** station data, 174 676 observations. This is why the campaign no longer depends on ERA5 for a soil claim.
Snow cover	**Rutgers** and Russian snow courses (14 369 Siberian DJF surveys).
Cloud amount	**CERES** and **MODIS**, which agree with each other, over dark ocean only — over ice they disagree by 7.7 pp.
Ocean T/S	**PHC3** on the *new* CORE3 mesh (220509 nodes). The 211567-node beta mesh climatology is a different file and mixing them fails silently.
Drake Passage	**Cunningham 2003** 134 ± 11.2 Sv and **Donohue 2016** 173.3 ± 10.7 Sv. Donohue includes near-bottom flow a 47-level mesh cannot carry, so the gap to 134 is the real bias.
Cavity basal melt	**Rignot 2013** ∼1500 and **Adusumilli 2020** ∼1100 $\mathrm{Gt\,yr^{-1}}$; the calibration target is 1100–1325.
Composite skill	**cmpitool** against ERA5, thirteen variables by region and season. Useful for regressions, blunt for mechanisms.
PI temperature	**ERA5 1990–2014 + HadCRUT5** warming 1850–1900 → 1990–2014 per 5° box (0.817 K global). Coverage is thin at the poles, and the model's own epoch warming differs (it under-warms Siberia).
Sea-ice drift	**OSI-455** v1.0 daily drift 2005–2014, satellite vectors only, with **NCEP-2** 10 m winds. A positive-SAM epoch, and satellite drift misses fast MIZ ice.
Labrador Sea ice	**HadISST2** March 1979–2008: 0.16 in the 56–62 N 60–50 W box, 0.37 over the basin; its pre-1979 values are a flat climatology.

How long an arm has to run. The standing threshold is $2\sigma\sqrt{2/N}$ from the control's own scatter in the same window, with no autocorrelation correction, so it is a floor rather than a true confidence interval — and branched arms are scored *paired*, because they share a restart and the unpaired form discards that and once hid a resolved result. Measured on this configuration: Drake Passage scatter ∼1.5 Sv; ice volume needs 20 years to resolve $\pm 1 \times 10^3$ $\mathrm{km^3}$/decade; decadal TOA means are stable to $\sim \pm 0.1$ $\mathrm{W\,m^{-2}}$; wallclock varies ±15 %, so SYPD comparisons need more than one leg each. The per-band one-year thresholds from round 26 remain in use: SO SW CRE annual ±1.97, SO DJF ±3.82, tropics annual ±0.99, global ±1.11.

The full per-window diagnostic set is `reval` (`release_evaluation_tool2`), driven by one config per window. Copying `configs/awiesm3_v35_pical_ccnice_2090s.py` and changing four fields (`model_version`, `pi_ctrl_start/end`, `spinup_end`, `out_path`) is the whole procedure; it submits 26 jobs and an HTML report.

22 Method notes

Techniques that earned their keep, recorded because they generalise.

Measure the reference-period offset, do not estimate it. Applying an *observed* epoch change to a model that does not reproduce that change put the boreal bias off by a factor of two and cost a withdrawn conclusion. The AMIP present-day run measures it directly.

Verify the artefact that executes, not the one that was edited. A namelist parameter is live only if the *staged* shared library carries it, and the file that is read may not be the one that looks live — LPJ-GUESS reads a hard-wired `guess.ins` while several plausible instruction files are staged beside it and never opened. Both traps cost a run each.

Prove a physical diagnostic against the input file's own metadata. The IDEMIX gate works because 0.9179869 TW is the forcing file's `Etot_in_TW_bin` attribute, not a remembered number. A self-consistency check would have passed on the broken build.

Use the model's own integrals where they exist. Hemispheric ice volume comes from FESOM's `sivoln/sivols`, computed on the native mesh inside the model, so it cannot pick up the wrong mesh. Likewise Drake Passage is computed on the model's own transect path rather than by interpolating to a straight line, which would not conserve.

Screen at one year, or offline, before spending forty. The basal-melt calibration was settled by three one-year runs; the snow scheme was validated offline before implementation; the forcing-transfer test used two spin-ups that already existed.

Keep one store, append to it. `data/coupled_annual_diag.nc` holds annual TOA, surface and four-band ocean heat content for 90 runs across 750 model years, and survives the deletion of the raw output it was built from. Several results here would be unrecoverable otherwise.

23 The software stack, and the move to albedo

The production runs of rounds 41–45 used the old FESOM build plus local branches (GM arms) or the new build (`ts1200`, `ihf*`, `nx1800`). For the last round a fresh `awiesm3-develop-cc` was installed on 2026–10–02/03 at `/albedo/work/user/jstreffi/model_codes/awiesm3-develop-cc` with `esm_master comp-awiesm3-develop-cc`:

component	branch	state
esm_tools	`feat/awiesm3-v3.4-co2`	`cf0a27943` carried the v3.5 CORE3 defaults, the mixing scheme and the IDEMIX forcing paths for the PICAL line.
fesom-2.8	`main` @ `82c67a4a`	Contains #1060, #1064 and #1087. Built coupled, double precision, `RECOM_COUPLED=ON`, XIOS on. Hemispheric GM (#1112) still to be added.
oifs-48r1	`awiesm3-develop` @ `f6d1860` (fork)	Five commits past develop's `872c4d2`: 2 m temperature to the ice sheet, and the surface-pressure mass fixer.
LPJ-GUESS	`lpj_guess_awiesm3` @ `179fb0d`	The AWI land-grid line with the canopy work.
OASIS, rnfmap, XIOS	`local_combined_fixes` @ `898c8b2`, `enthalpy_on_ocean_side` @ `a635cc6`, `main` @ `9c33506d`	XIOS re-cloned.

Merged upstream during the campaign: FESOM PR #1054 (the single-precision ocean heat leak), #1064 (`cavity_gamma_scale`, `f3c3d3fa`), #1060 (ice skin from the atmosphere, `57c433e5`), #1087 (the two CVMix bugs, `18926b72`), and three CI fixes of our own that unblocked stacked pull requests in that repository (#1088, #1089, #1091). ~~Still open and needed: #1060 and #1087.~~ Both are merged. Open: FESOM #1103 (IDEMIX smoothing per unit time), #1112 (hemispheric GM), and `release_evaluation_tool2` #7 (the Gregory snow-enthalpy fix and seven other albedo fixes).

On albedo the 1800 s production layout is 768 FESOM ranks at 96 per node on 25 nodes; the 1200 s step needs 33. As of 2026–09–26 the campaign was *moving* to albedo for queue time: a 47-node leg on levante had waited between 30 seconds and 10 hours to start. The branch point is 2100, the only year with a complete consistent restart set. Three one-liners kill a first run there, all in `HANDOFF_albedo_spinup_continuation.md`: `h0min` in the runscript aborts FESOM at the namelist read on any branch except the abandoned `feat/leadclose-h0min`; both IDEMIX forcing files must exist under `${fesom.pool_dir}/forcing/idemix/`; and `ltos` must go in `ecearth.ins.j2`, the file the run actually reads. The standing gate on any IDEMIX leg is `Etot_srf` 0.3211645 and `Etot_bot` 0.9135067 → 0.9179869 TW within the first minute of the compute log; the failure signature is 6.05×10^{-4} with -1.54×10^{16}, which NaN'd the ocean in 100 seconds on 2026–09–24. Two later notes: round 45 found that the `/ice_therm/` read does not check `iostat`, so an unknown name there ends the read silently rather than aborting (the abort above was not re-examined against this); and the pool IDEMIX forcing has been unreadable since 2026–10–01, so the runscripts point `idemix_botforc_file` at a project-local copy. The spin-up plots of this summary come from `release_evaluation_tool2` branch `feat/critical-path-spinup` (per-year cache, annotations, AMOC series).

The release (2026–10–07). The spin-up runs on tags only: FESOM `2.8.2` (adds the pressure coupling, PR #1115), OpenIFS `48r1v6` (`fc271eb`: pressure coupling, DMS namelist fix, sea-ice roughness entries), LPJ-GUESS `4.1.13`, OASIS `5.3`, XIOS `2.5.4`, runoff mapper `1.4`; `esm_tools v3.5.0` updated in place, its tag last. Development continues on `develop`, the basis for v3.5.1.

24 The path forward

As of 2026–10–15. Run the CMIP7 spin-up until ocean and land together drift by less than 0.1 GtC yr^{-1}, then branch piControl, and after 100 years of it historical, 1pctCO2 and abrupt-4xCO2; the scenarios follow historical. Decide whether the colder state is accepted before piControl starts, and document the structural biases (Southern Ocean cloud and cap, the diffuse Antarctic pack, the stratocumulus coasts, the warm, dry NH summer land, the abyssal drift).

The plan of 2026–10–03, kept for the record. Item 0 ran (rounds 46–47) and decided the mixing scheme; item 1 (NH GM 1500) was not run; item 2 became the spin-up configuration.

0. Added the same day: TKE without IDEMIX. PICAL_crunveg_gmhemi_tke, 2140–2159 from gmhemi1800, one variable changed, runs first (§6). Score it on heat by depth, both hemispheres' ice volume and net TOA before choosing the mixing scheme of the release.

1. Hemispheric GM with NH `K_GM_max` 1500. gmhemi1800 is right in everything but the Labrador Sea, which it overshoots (ice 0.10–0.13 against 0.16). A higher NH value is the direct correction, and the GM 1500 arm at 1200 s already showed a Labrador Sea that convects without capping.

2. Continue the current tuning on the `-cc` configuration. FESOM and LPJ-GUESS warm start, REcoM on in concentration-driven PI mode, OpenIFS cold start, on the fresh install of §23. Hemispheric GM must be carried into that build. *Done for one leg:* **PICAL__cc** ran 2140–2144 and is paused at a complete restart for performance work; it costs 6.7 times the physics-only line.

3. Document the structural biases of v3.5. The Southern Ocean cloud deficit, the diffuse Antarctic pack and its over-responsive drift, the warm stratocumulus coasts, and the warm, dry NH summer land. Each is measured, each resisted the levers aimed at it or has no physical lever left, and each is better stated plainly than tuned around.

Note what is *not* on this list. DMS / marine CCN was reconsidered as a global cooling lever and declined; `h0min` is not a lever in the coupled build; the McPhee ice–ocean flux went the wrong way in both hemispheres; slowing the Antarctic drift would likely cost SH extent. The atmospheric lever families were, by measurement, nearly exhausted before these rounds.

25 Assets

what	where
Full record, 183 pages	`report/report.tex`
Previous summary (2026–08–24)	`report/summary_superseded_20260926.tex`; the 2026–09–26 version of this one is in git history
Albedo handoff	`HANDOFF_albedo_spinup_continuation.md`
Annual TOA / surface / OHC store	`data/coupled_annual_diag.nc`, built by `scripts/analysis/coupled_annual_store.py`
Drift, ice volume, GIOMAS	`scripts/analysis/ccnice_regional_drift.py`, `ice_volume_series.py`, `giomas_volume_targets.py` (takes `OBS_Y0`/`OBS_Y1`)
Drake Passage, cavity melt	`scripts/analysis/drake_passage_transport.py` and `scripts/figures/cavity_melt_maps_mixing.py`; both need the conda env `esm-tools_auto_tripyview`, and the transport needs `PYTHONPATH=/work/ab0246/a270092/software/tripyview`
Run and parameter notes	`notes/RUNS_AND_PARAMETERS.md`
Labrador history and noise floor	`scripts/analysis/labrador_any_mesh.py`, `labrador_ice_vs_global_sst.py`, `labrador_mld_noise_floor.py`
GM arms and heat pathways	`scripts/analysis/gm_arms_global.py`, `heat_pathways.py`, `gm_k_rossby.py`; `scripts/figures/gm_k_maps.py`; tables `data/clim/gm_arms1800_*.txt`
PI bias and warm regions	`scripts/analysis/pi_t2m_bias_epoch_corrected.py`, `pi_t2m_bias_land_ocean.py`, `warm_regions_diag.py`; `scripts/figures/pi_t2m_bias_maps.py`
Sea-ice budget and drift	`scripts/analysis/ice_winter_budget_hemispheres.py` (env `SIC`), `ice_drift_vs_osisaf.py`, `ice_wind_response.py`, `ice_drag_effective.py`; obs under `/albedo/work/user/jstreffi/obs/`
Full `reval` diagnostics	`/work/ab0246/a270092/software/release_evaluation_tool2`, per-window configs under `configs/`
Configuration	esm_tools `feat/awiesm3-v3.4-co2` at `cf0a27943`; the release checklist `notes/v3.5.0_final_tag_round.md`
Arms of rounds 53–55	`scripts/analysis/overnight_arms_first_decade.py`, `overnight_arms_volumes_drift.py`, `ice_speed_and_so_convection.py`; tables `data/clim/arms_with_ctl2.txt`, `icb_four_decades.txt`, `z0_two_decades.txt`
Iceberg scheme	esm_tools `couplings/fesom/iceberg_budget.py` and the `icb_seed` / `icb_budget` jobs in `awiesm3.yaml`
Spin-up evaluation	`scripts/analysis/spinup_v350_annual_series.py`, `spinup_v350_hovmoeller_deep.py`, `deep_warming_where.py`; `reval` part 27 (carbon pools); `plots/reval_spinup_v350cc_*`